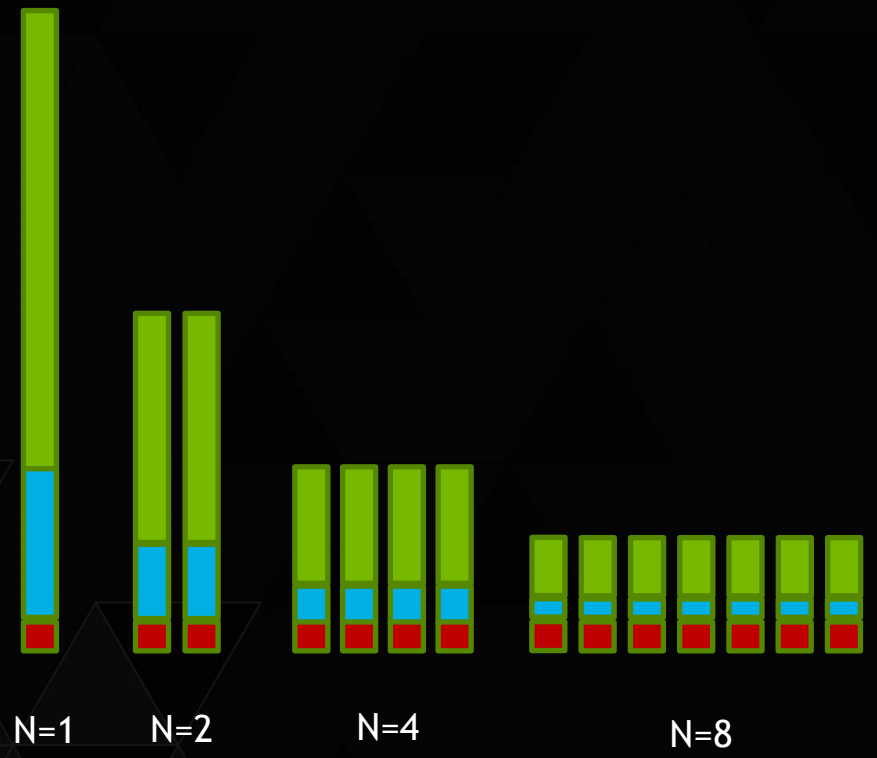
GPU TECHNOLOGY CONFERENCE

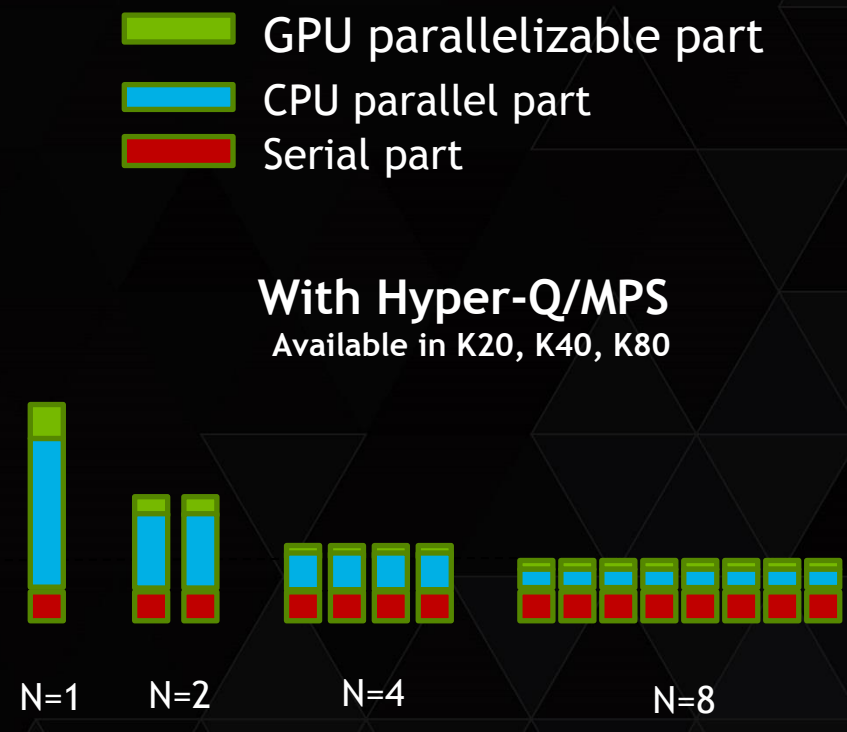
IMPROVING GPU UTILIZATION WITH MULTI-PROCESS SERVICE (MPS)

PRIYANKA,
COMPUTE DEVTECH, NVIDIA

STRONG SCALING OF MPI APPLICATION



Multicore CPU only



GPU accelerated CPU

WHAT YOU WILL LEARN

- Multi-Process Server
- Architecture change (HyperQ - MPS)
- MPS implication on Performance
- Efficiently utilization of GPU under MPS
- Profile and Timeline
- Example

WHAT IS MPS

- CUDA MPS is a feature that allows multiple CUDA processes to share a single GPU context. each process receive some subset of the available connections to that GPU.
- MPS allows overlapping of kernel and memcopy operations from different processes on the GPU to achieve maximum utilization.
- Hardware Changes - Hyper-Q which allows CUDA kernels to be processed concurrently on the same GPU

REQUIREMENT

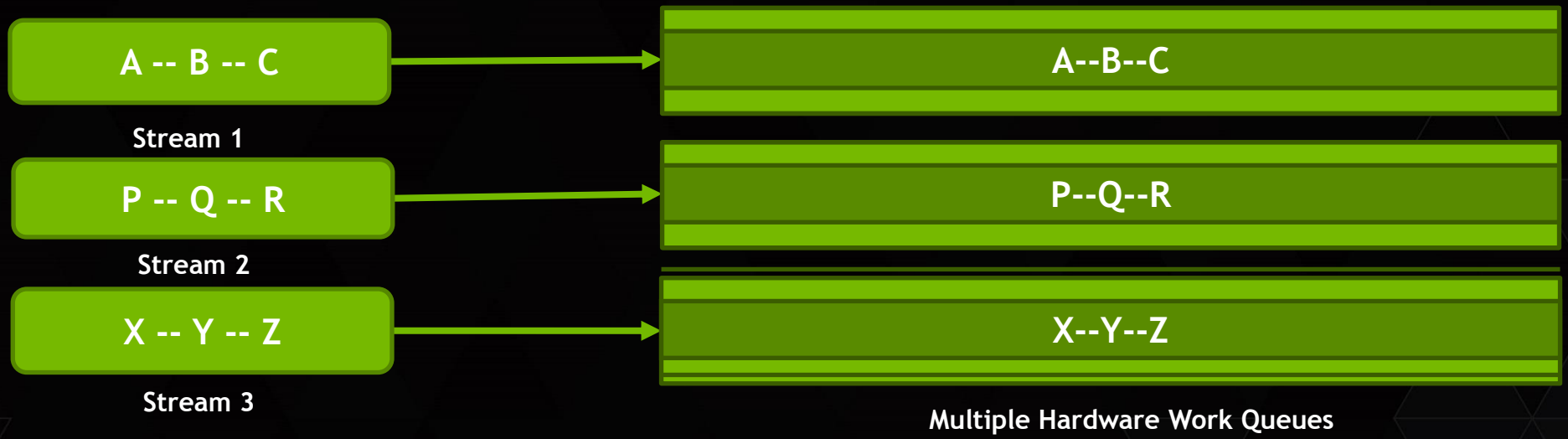
- Supported on Linux
- Unified Virtual Addressing
- Tesla with compute capability version 3.5 or higher, Toolkit - CUDA 5.5 or higher
- Exclusive-mode restrictions are applied to the MPS server, not MPS clients

ARCHITECTURAL CHANGE TO ALLOW THIS FEATURE

CONCURRENT KERNELS

- GPU can run multiple independent kernels concurrently
  - Fermi and later (CC 2.0)
  - Kernels must be launched to different streams
  - Must be enough resources remaining while one kernel is running
- While kernel A runs, GPU can launch blocks from kernel B if there are sufficient free resources on any SM for at least one B block
  - Registers, shared memory, thread block slots, etc.
- Max concurrency: 16 kernels on Fermi, 32 on Kepler
  - Fermi further limited by narrow stream pipe…

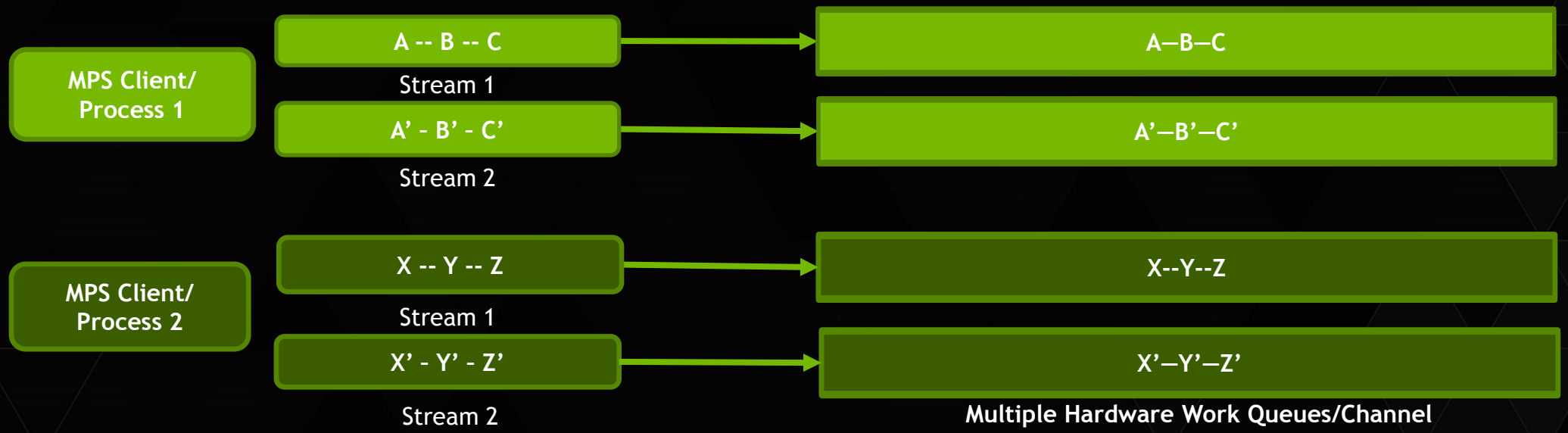
KEPLER IMPROVED CONCURRENCY



Kepler allows 32-way concurrency

- One work queue per stream
- Concurrency at full-stream level
- No inter-stream dependencies

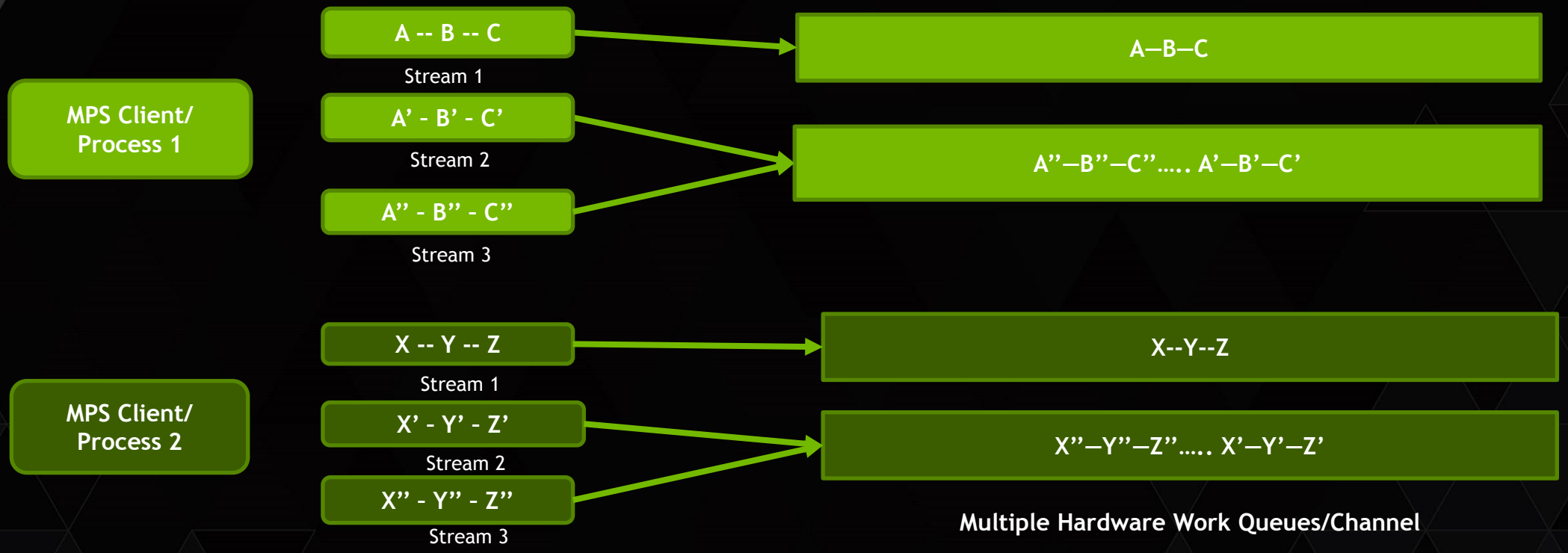
CONCURRENCY UNDER MPS



Kepler allows 32-way concurrency

- One work queue per stream, 2 work queue per MPS Client
- Concurrency at 2 stream level per MPS client, total 32
  - Case 1: N_stream per MPS Client< N_channel (i.e. 2), - no serialization

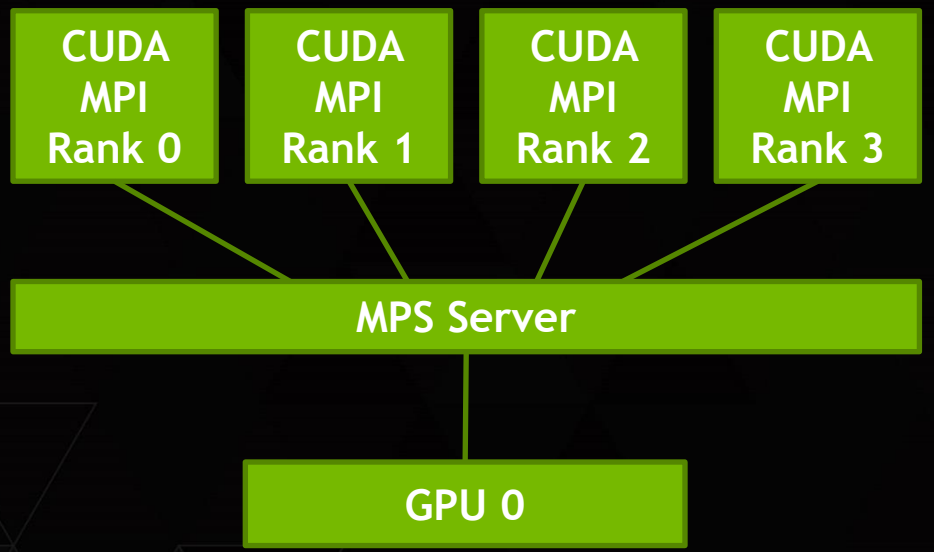
SERIALIZATION/FALSE DEPEDENCY UNDER MPS



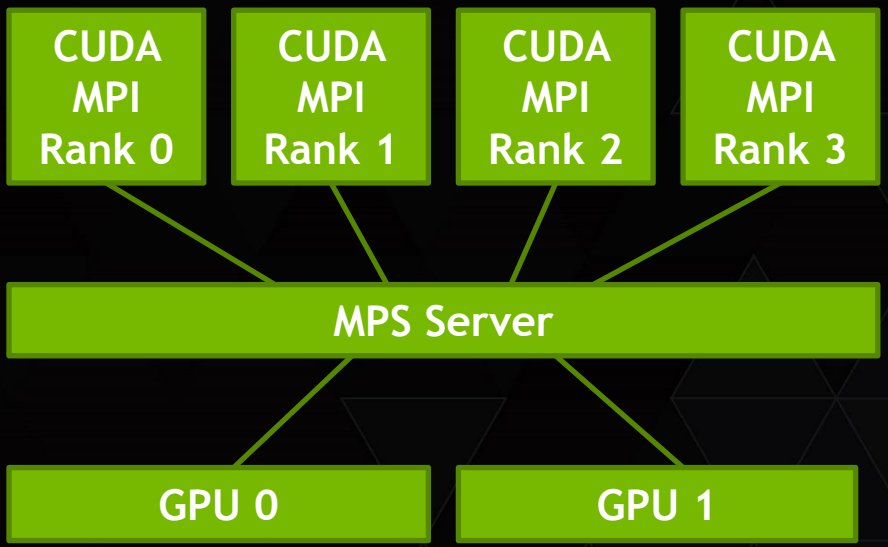
Kepler allows 32-way concurrency

- One work queue per stream, 2 work queue per MPS Client
- Concurrency at 2 stream level per MPS client, total 32
  - Case 2: N_stream>N_channel - False dependency/serialization

HYPER Q/MPI (MPS): SINGLE/MULTIPLE GPUS PER NODE



MPS Server efficiently overlaps work from multiple ranks to single GPU



MPS Server efficiently overlaps work from multiple ranks to each GPU

Note : MPS does not automatically distribute work across the different GPUs. Inside the application user has to take care of GPU affinity for different mpi rank .

HOW MPS WORK

Process 1 initiated before MPS Server started

MPS Server

Many to one context mapping

MPS Client

MPI Process 2 -
Create CUDA context
MPI Process 2 -
Create CUDA context

All MPS Client Process started after starting MPS server will communicate through MPS server only

Allows multiple CUDA processes to share a single GPU context

HOW TO USE MPS ON SINGLE GPU

- No application modifications necessary
- Proxy process between user processes and GPU
- MPS control daemon
  - Spawn MPS server upon CUDA application startup
- Setting
  - **export CUDA_VISIBLE_DEVICES=0**
  - **nvidia-smi -i 0 -c EXCLUSIVE_PROCESS**
  - **nvidia-cuda-mps-control -d**
- Enabled via environment variable (for CRAY)
  export CRAY_CUDA_PROXY=1

USING MPS ON MULTI-GPU SYSTEMS

Step 1 : Set the GPU in exclusive mode

- sudo nvidia-smi -c 3 -i 0,1

Step 2 : Start the mps deamon (In first window) & Adjust pipe/log directory

- export CUDA_VISIBLE_DEVICES= ${DEVICE}
- export CUDA_MPS_PIPE_DIRECTORY=${HOME}/mps${DEVICE}/pipe
- export CUDA_MPS_LOG_DIRECTORY=${HOME}/mps${DEVICE}/log
- nvidia-cuda-mps-control -d

Not required in CUDA 7.0

Step 3 : Run the application (In second window)

- Mpirun -np 4 ./mps_script.sh
- NGPU=2
- lrank=$MV2_COMM_WORLD_LOCAL_RANK
- GPUID=$(($lrank%$NGPU))
- export CUDA_MPS_PIPE_DIRECTORY=${HOME}/mps${DEVICE}/pipe

(for MV2_COMM_WORLD_LOCAL_RANK for mvapich2, OMPI_COMM_WORLD_LOCAL_RANK for openmpi)

- **Step 4 : Profile the application (if you want to profile your mps code)**
  - nvprof -o profiler_mps_mgpu$lrank.pdm ./application_exe

NEW IN CUDA 7.0

Step 1 : Set the GPU in exclusive mode

```
sudo nvidia-smi  -c 3 -i 0,1
```

Step 2 : Start the mps deamon (In first window) & Adjust pipe/log directory

```
export CUDA_VISIBLE_DEVICES= ${DEVICE}
nvidia-cuda-mps-control -d
```

Step 3 : Run the application (In second window)

```
lrank=$OMPI_COMM_WORLD_LOCAL_RANK
case ${lrank} in
[0]) export CUDA_VISIBLE_DEVICES=0; numactl —cpunodebind=0 ./executable;;
[1]) export CUDA_VISIBLE_DEVICES=1; numactl —cpunodebind=1 ./executable;;
[2]) export CUDA_VISIBLE_DEVICES=0; numactl —cpunodebind=0 ./executable;;
[3]) export CUDA_VISIBLE_DEVICES=1; numactl —cpunodebind=1 ./executable;
esac
```

GPU UTILIZATION AND MONITORING MPI PROCESS RUNNING UNDER MPS OR WITHOUT MPS

GPU Utilization by different MPI Rank Without MPS

Two MPI Rank per processor sharing same GPU

```
[psah@ivb111 ~]$ nvidia-smi
Thu Feb 26 00:46:13 2015
+------------------------------------------------------+
| NVIDIA-SMI 346.29     Driver Version: 346.29         |
|-------------------------------+----------------------+----------------------+
| GPU  Name        Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|         Memory-Usage | GPU-Util  Compute M. |
|===============================+======================+======================|
|   0  Tesla K40m          On   | 0000:04:00.0     Off |                    0 |
| N/A   37C    P0    76W / 235W |    678MiB / 11519MiB |     48%      Default |
+-------------------------------+----------------------+----------------------+
|   1  Tesla K40m          On   | 0000:05:00.0     Off |                    0 |
| N/A   34C    P0    78W / 235W |    677MiB / 11519MiB |     39%      Default |
+-------------------------------+----------------------+----------------------+
|   2  Tesla K40m          On   | 0000:08:00.0     Off |                    0 |
| N/A   34C    P8    31W / 235W |     56MiB / 11519MiB |      0%      Default |
+-------------------------------+----------------------+----------------------+
|   3  Tesla K40m          On   | 0000:09:00.0     Off |                    0 |
| N/A   34C    P8    31W / 235W |     56MiB / 11519MiB |      0%      Default |
+-------------------------------+----------------------+----------------------+
|   4  Tesla K40m          On   | 0000:83:00.0     Off |                    0 |
| N/A   34C    P8    32W / 235W |     56MiB / 11519MiB |      0%      Default |
+-------------------------------+----------------------+----------------------+
|   5  Tesla K40m          On   | 0000:84:00.0     Off |                    0 |
| N/A   37C    P8    33W / 235W |     56MiB / 11519MiB |      0%      Default |
+-------------------------------+----------------------+----------------------+

+-----------------------------------------------------------------------------+
| Processes:                                                       GPU Memory |
|  GPU       PID  Type  Process name                               Usage      |
|=============================================================================|
|    0     40764    C   ./test_real2                                 309MiB |
|    0     40767    C   ./test_real2                                 309MiB |
|    1     40765    C   ./test_real2                                 309MiB |
|    1     40766    C   ./test_real2                                 309MiB |
+-----------------------------------------------------------------------------+
```

GPU Utilization by different MPI Rank under MPS

```
[psah@ivb193 ~]$ nvidia-smi
Thu Feb 26 02:10:19 2015
+------------------------------------------------------+
| NVIDIA-SMI 346.29     Driver Version: 346.29         |
|-------------------------------+----------------------+----------------------+
| GPU  Name        Persistence-M| Bus-Id        Disp.A | Volatile Unc      ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|         Memory-Usage | GPU-Util  Co ute M.  |
|===============================+======================+======================|
|   0  Tesla K20m          On   | 0000:04:00.0     Off |                    0 |
| N/A   40C    P0   114W / 225W |   1106MiB /  4799MiB |     55%   E. Process |
+-------------------------------+----------------------+----------------------+
|   1  Tesla K20m          On   | 0000:05:00.0     Off |                    0 |
| N/A   40C    P0   116W / 225W |   1105MiB /  4799MiB |     72%   E. Process |
+-------------------------------+----------------------+----------------------+
|   2  Tesla K20m          On   | 0000:83:00.0     Off |                    0 |
| N/A   35C    P8    26W / 225W |     14MiB /  4799MiB |      0%      Default |
+-------------------------------+----------------------+----------------------+
|   3  Tesla K20m          On   | 0000:84:00.0     Off |                    0 |
| N/A   35C    P8    26W / 225W |     14MiB /  4799MiB |      0%      Default |
+-------------------------------+----------------------+----------------------+

+-----------------------------------------------------------------------------+
| Processes:                                                       GPU Memory |
|  GPU       PID  Type  Process name                               Usage      |
|=============================================================================|
|    0     33925    C   nvidia-cuda-mps-server                       1090MiB |
|    1     33924    C   nvidia-cuda-mps-server                       1089MiB |
+-----------------------------------------------------------------------------+
```

MPS PROFILING WITH NVPROF

Step 1: Launch MPS daemon

- $ nvidia-cuda-mps-control -d

Step 2: Run nvprof with --profile-all-processes

- $ nvprof --profile-all-processes -o apllication_exe_%p
- ======== Profiling all processes launched by user “user1"
- ======== Type "Ctrl-c" to exit

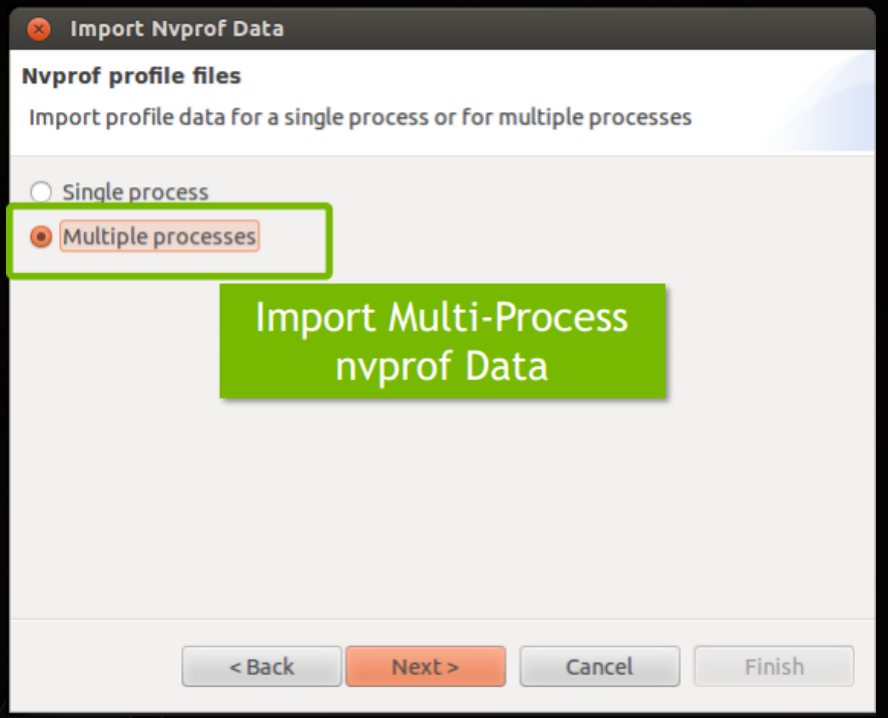
Step 3: Run application in different terminal normally

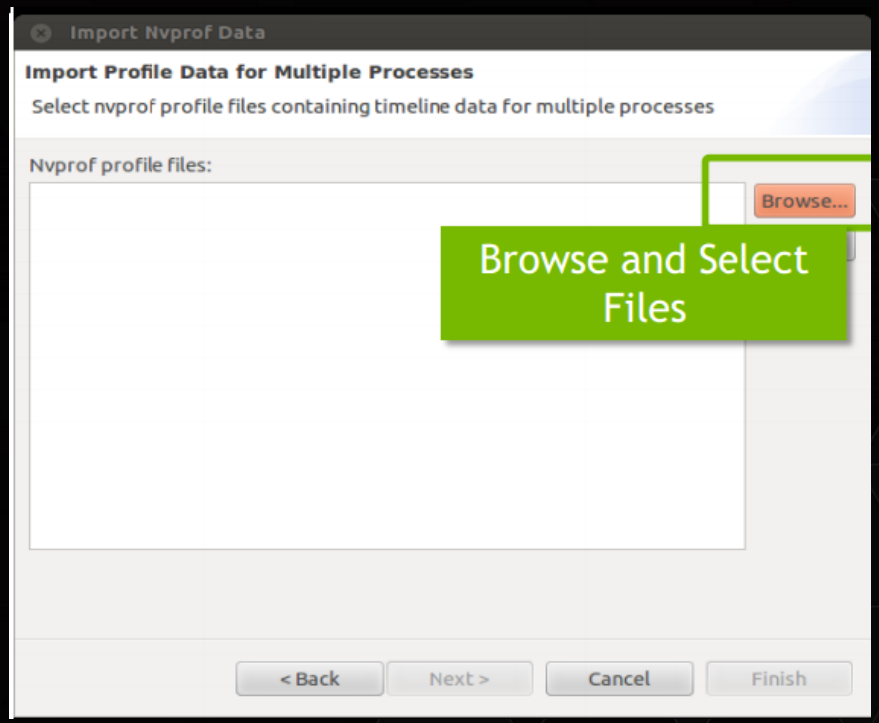
- $ application_exe

Step 4: Exit nvprof by typing Ctrl+c

- ==5844== NVPROF is profiling process 5844, command: application_exe
- ==5840== NVPROF is profiling process 5840, command: application_exe…
- ==5844== Generated result file: /home/mps/r6.0/application_exe_5844
- ==5840== Generated result file: /home/mps/r6.0/application_exe_5840

VIEW MPS TIMELINE IN VISUAL PROFILER

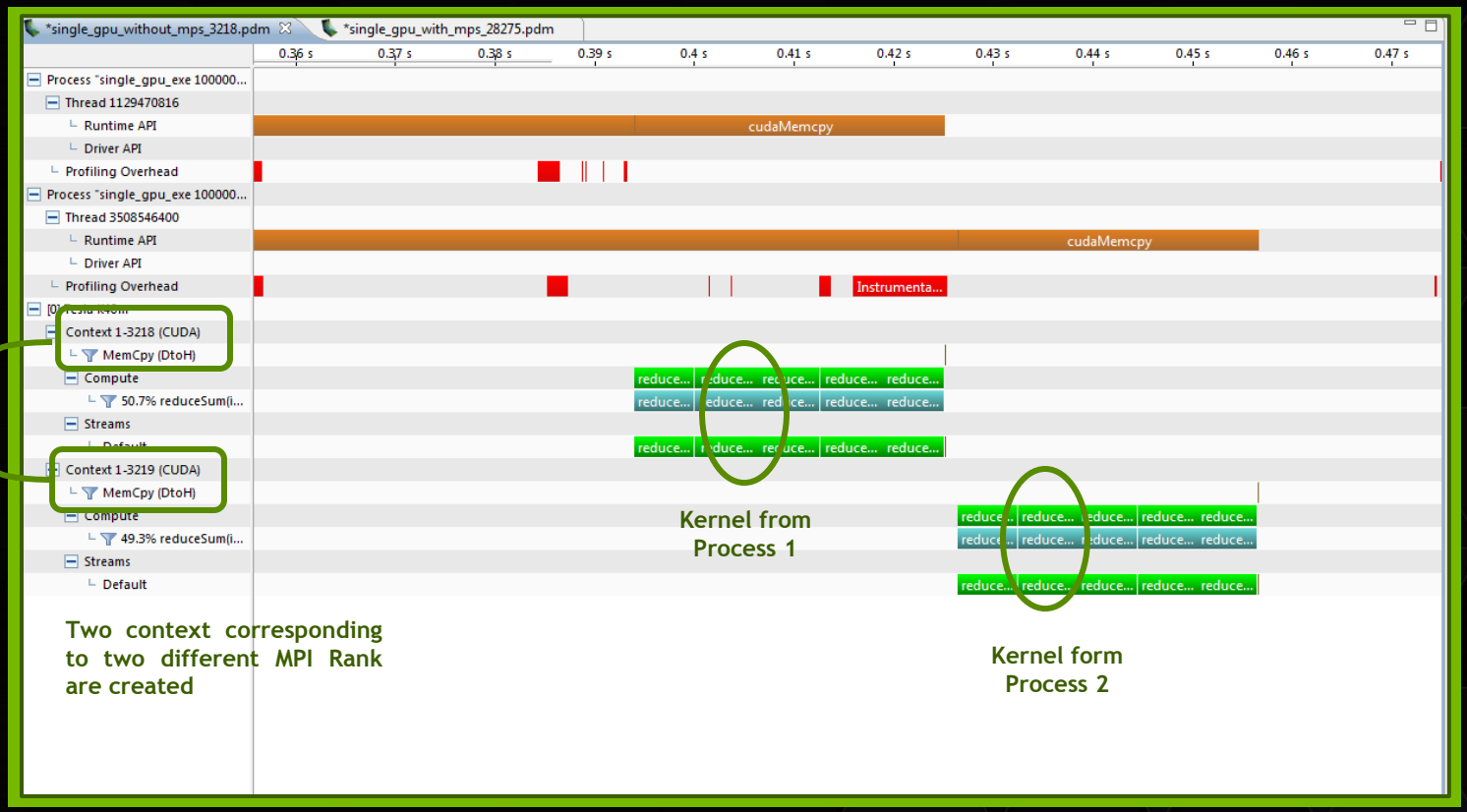




PROCESS SHARING SINGLE GPU WITHOUT MPS: NO OVERLAP

Process 1 -
Create CUDA context
Process 2 -
Create CUDA context

Allows multiple processes to create their separate GPU context



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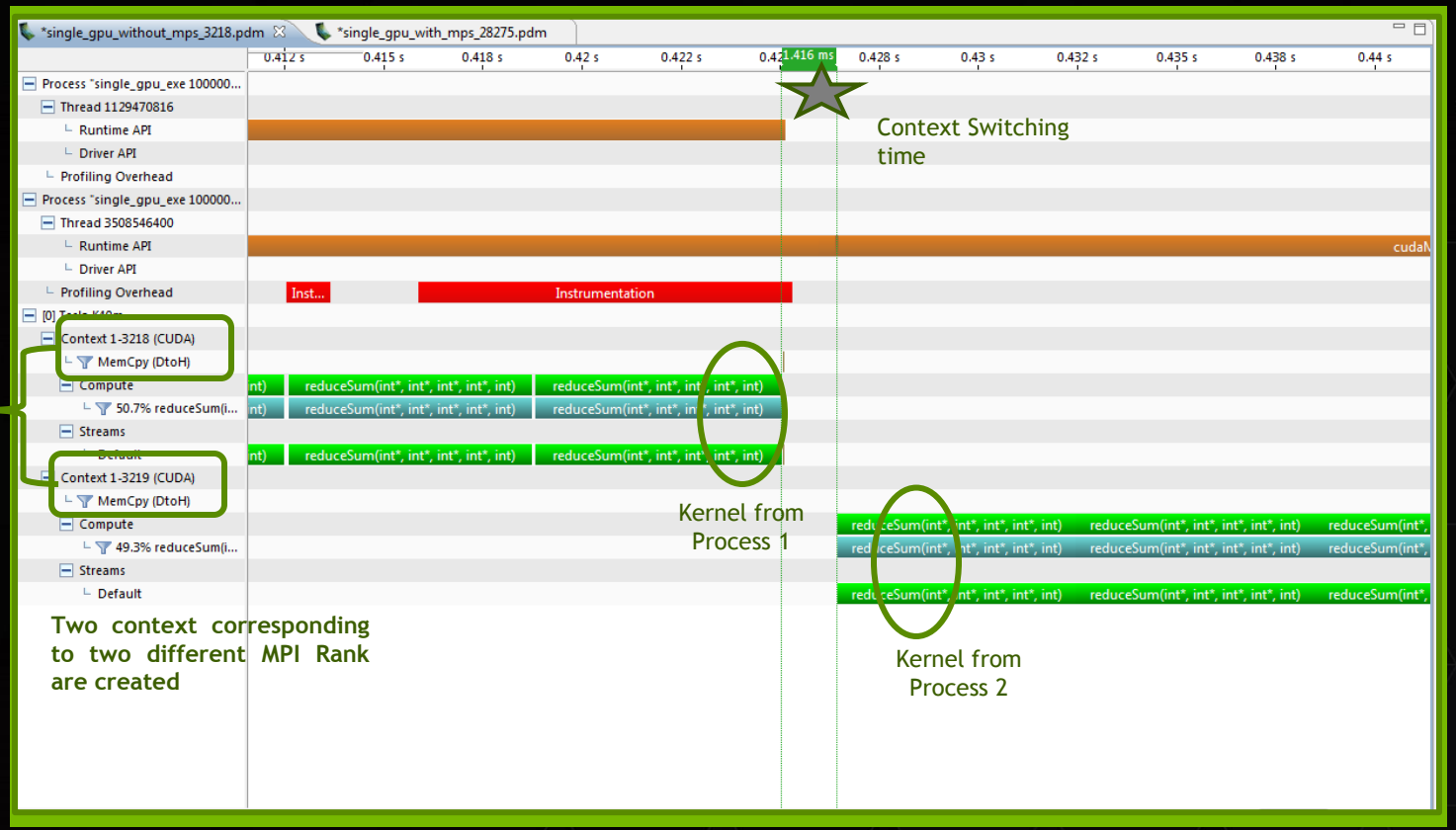
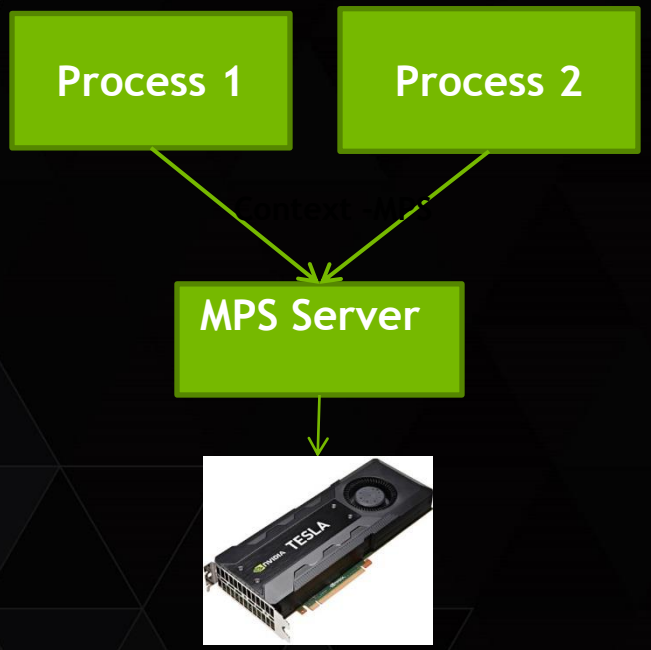
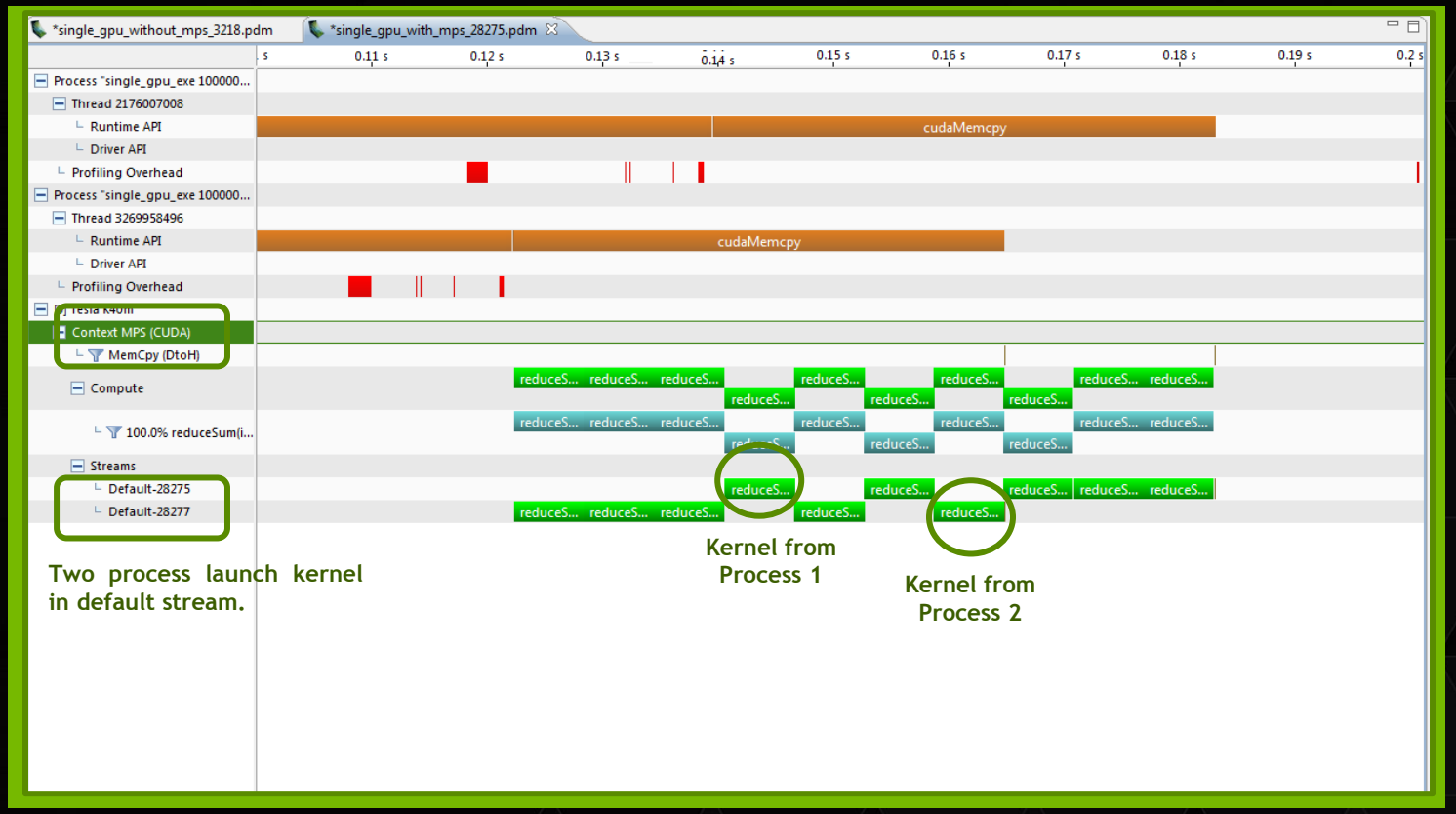
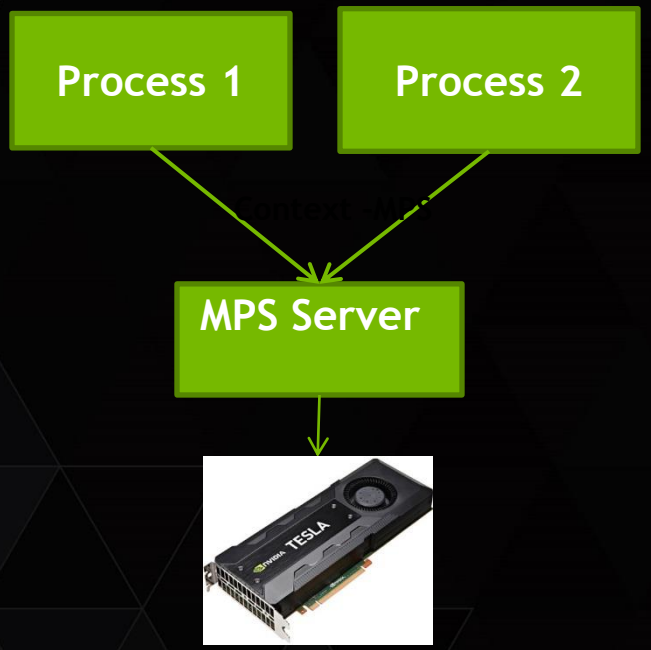


PROCESS SHARING SINGLE GPU WITH MPS: OVERLAP

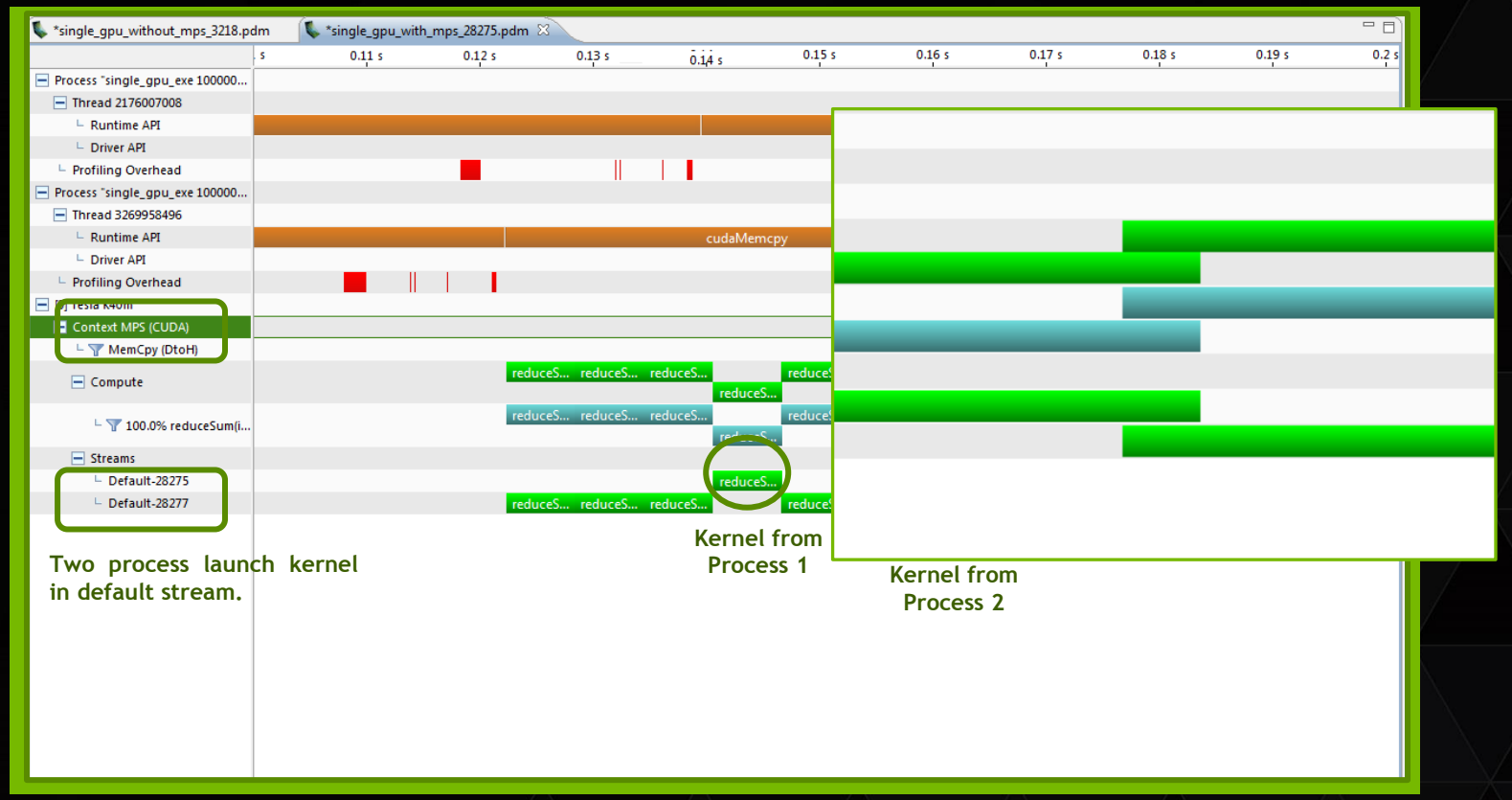




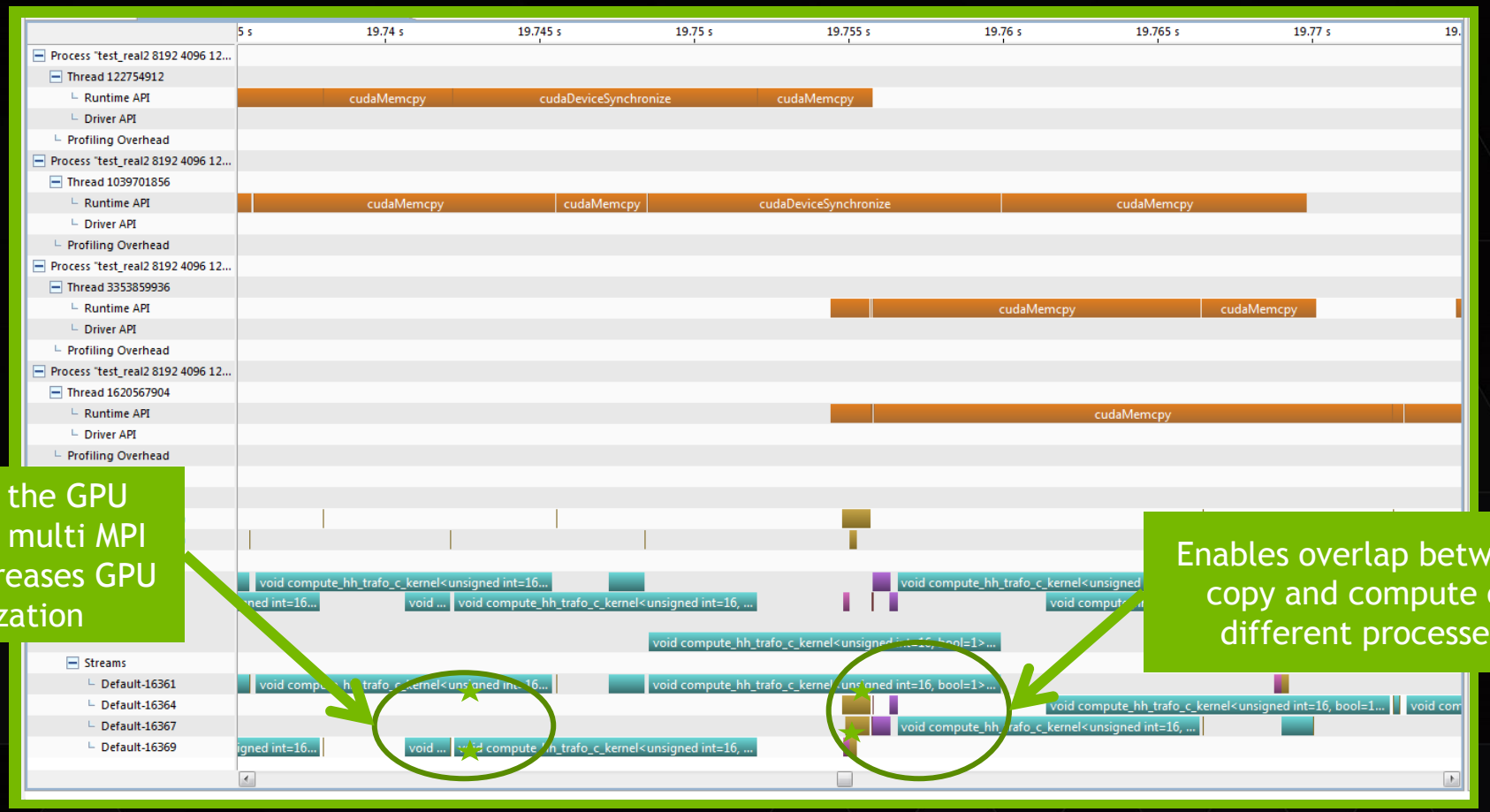
PROCESS SHARING SINGLE GPU WITH MPS: OVERLAP



Allows multiple processes to share single CUDA Context



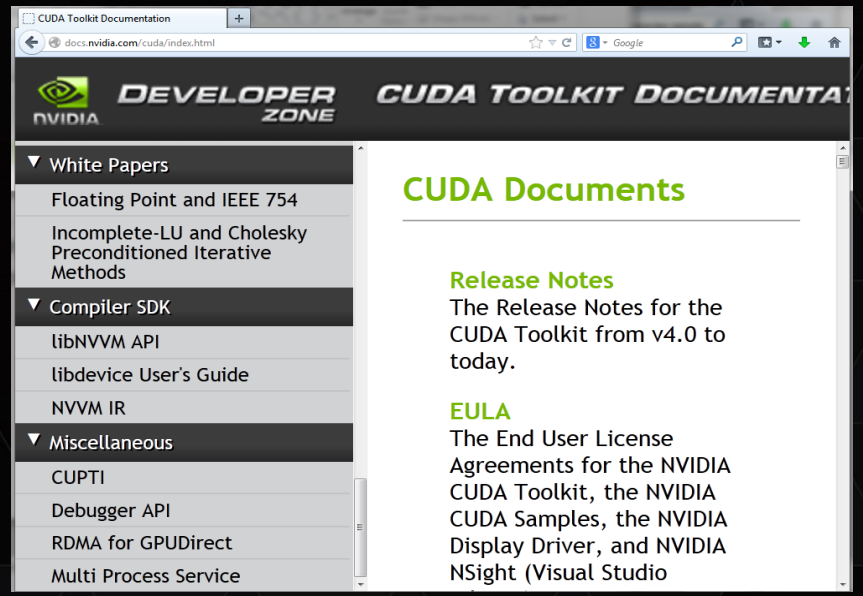
MULTIPLE PROCESS SHARING SINGLE GPU



REFERENCE

- S5117_JiriKraus_Multi_GPU_Programming_with_MPI

- Blog post by Peter Messmer of NVIDIA - http://blogs.nvidia.com/blog/2012/08/23/unleash-legacy-mpi-codes-with-keplers-hyper-q/



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Please complete the Presenter Evaluation sent to you by email or through the GTC Mobile App. Your feedback is important!

Warped-Slicer: Efficient Intra-SM Slicing through Dynamic Resource Partitioning for GPU Multiprogramming

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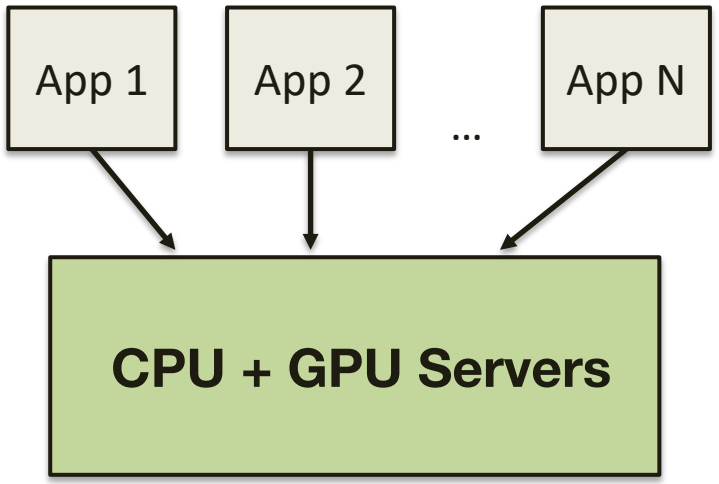
*Murali Annavaram**

** University of Southern California,* *San Jose State University,* ❖ *Yonsei University*

USC School of Engineering

University of Southern California

GPU Accelerated Computing Platforms

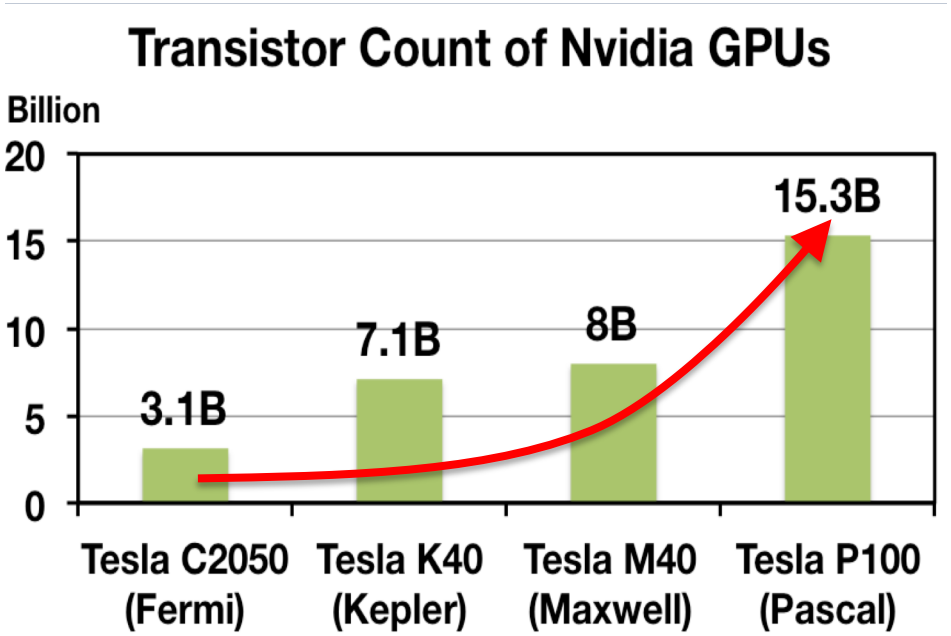


- Many vendors now providing cloud computing for GPU services
- **Multiple applications need to run on the same server**

[1] GTC 2016, Jen-hsun Huang, GPU-Accelerated Computing Changing the World

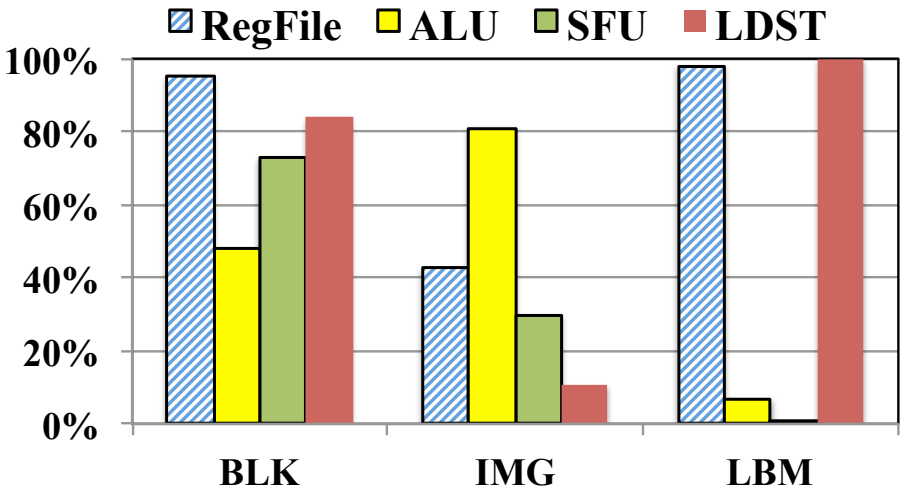
Ever-increasing amount of GPU resources

- Increasing number of execution resources in each GPU
- Resource underutilization is a growing problem
- Share resources across multiple kernels

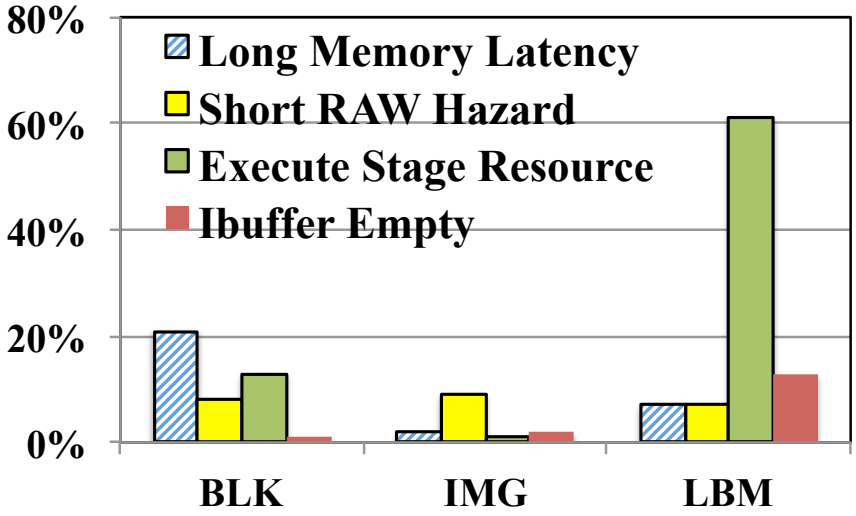


Resource usage variation across applications

- Utilization imbalance of register file and execution units



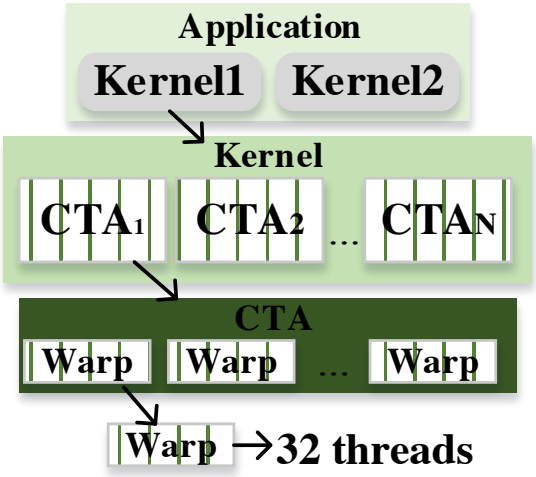
- Breakdown of pipeline stalls due to various reasons

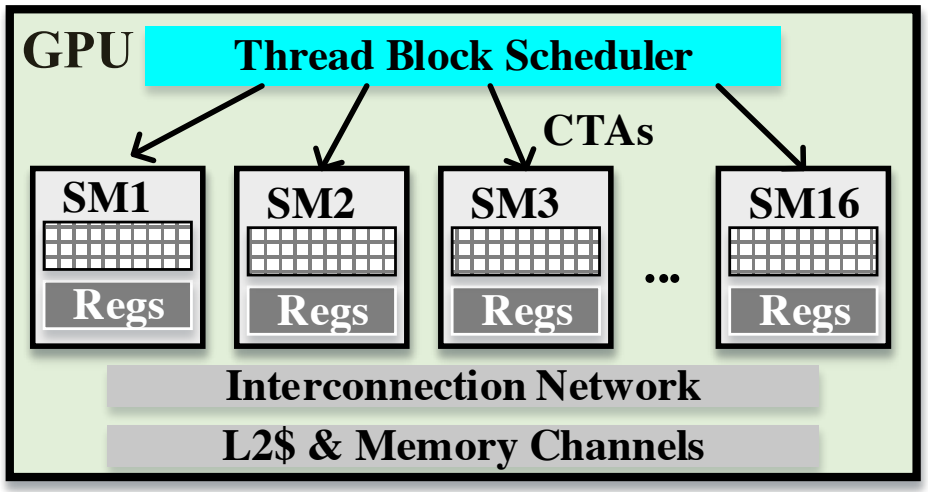


Idea: Exploit diverse application characteristics to alleviate resource imbalance

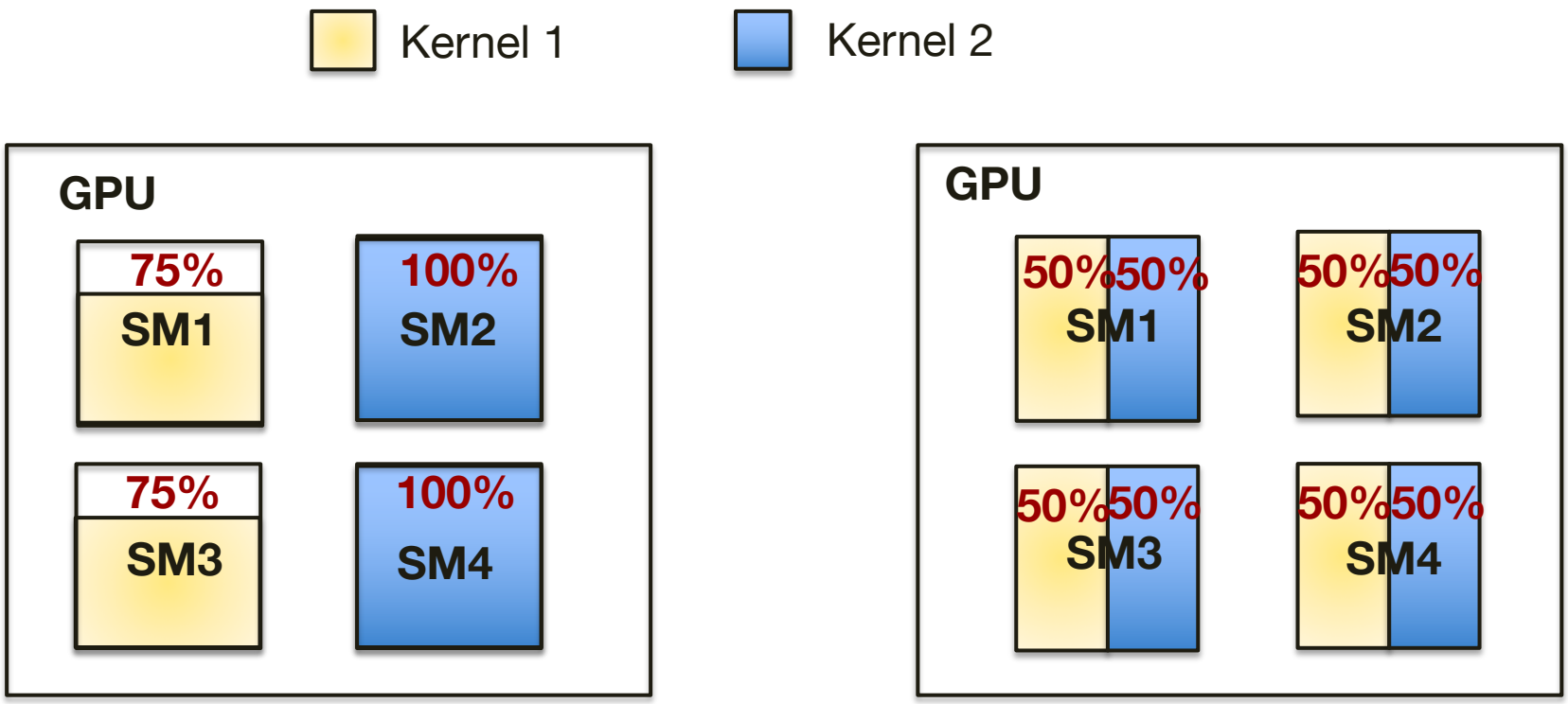
GPU Background

CTA: Cooperative Thread Array
a.k.a. thread block



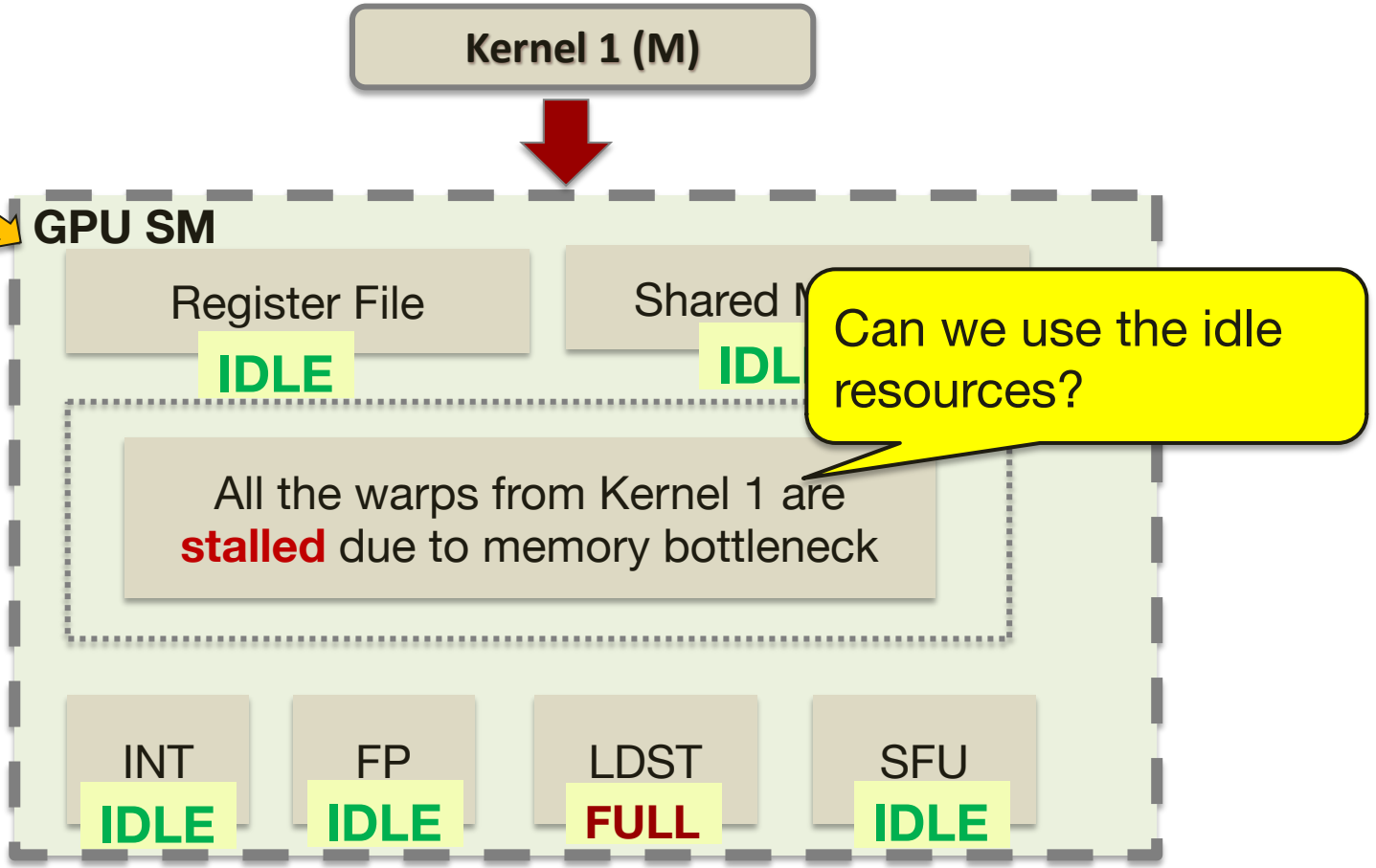


Previous work: spatial multitasking[1]

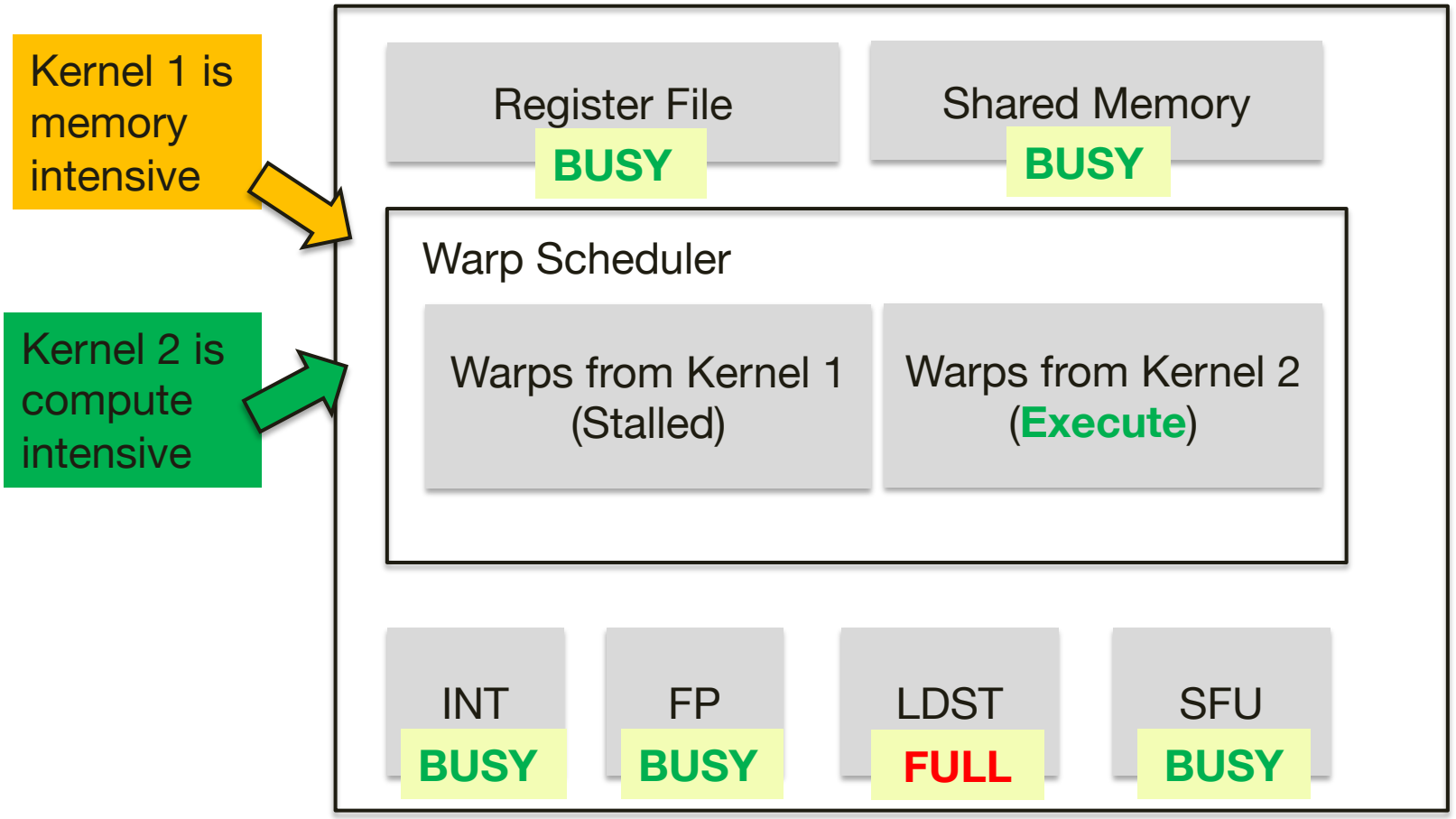


[1] J. T. Adriaens, K. Compton, N.S.Kim and M.J.Schulte, "The case for GPGPU spatial multitasking," HPCA, 2012

Example: a memory intensive kernel causes underutilization inside SM



Warped-Slicer: collocate with a compute intensive kernel



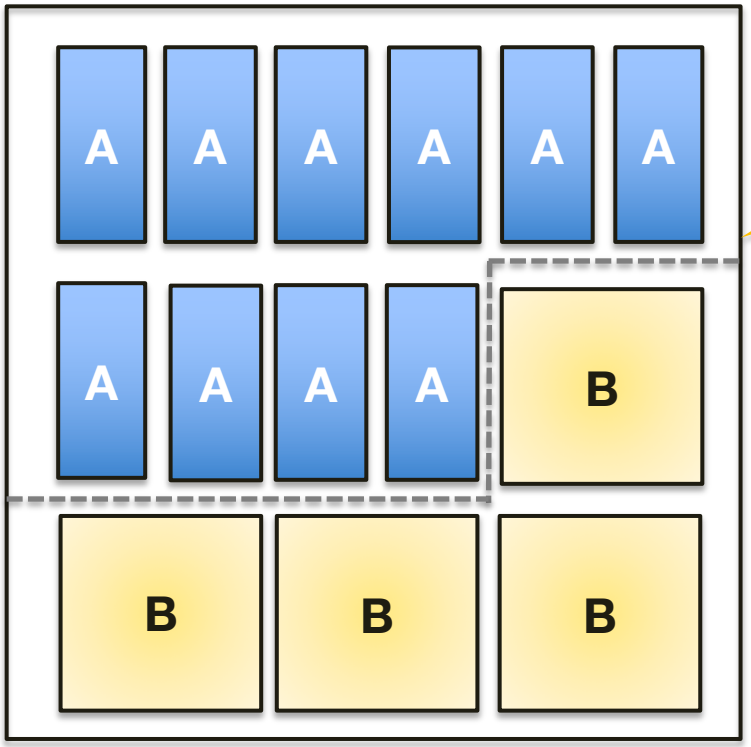
Intra-SM slicing strategies

- **Baseline** allocation strategy:
  - State-of-the-art GPU provides basic hardware support for multi-kernel co-location in the same SM
  - Follows a ***leftover allocation*** policy

- Another reasonable comparison point:
  - Even partitioning

- Our main contribution:
  - ***Warped-slicer*** – a *more efficient resource sharing technique than the above policies*

Warped-Slicer: an efficient resource partitioning

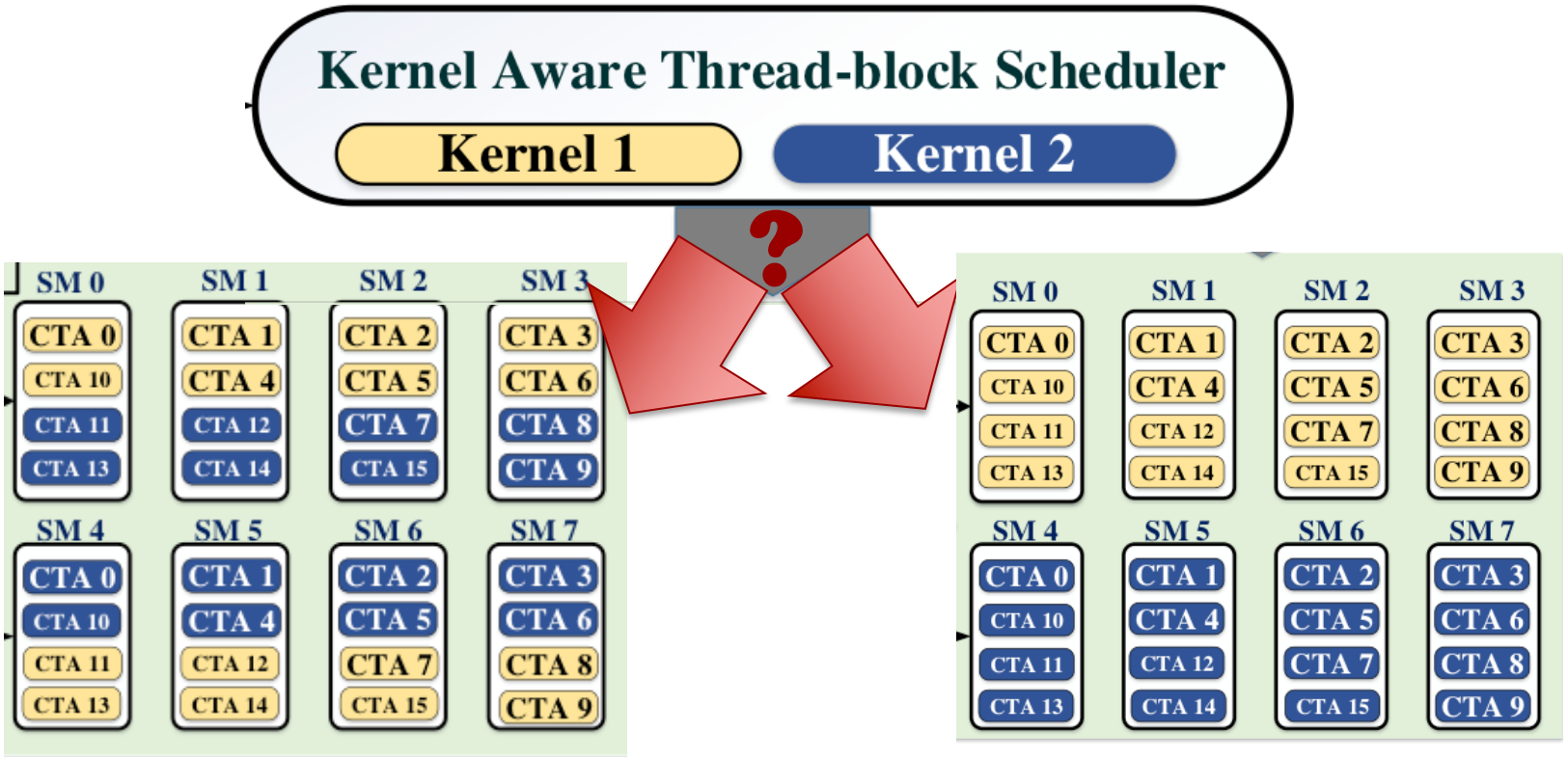


Different from Even partitioning, warped-slicer does ‘uneven’ partitioning

We can assign more register files to kernel A, while giving more shared memory to kernel B to maximize usage

Warped-slicer techniques

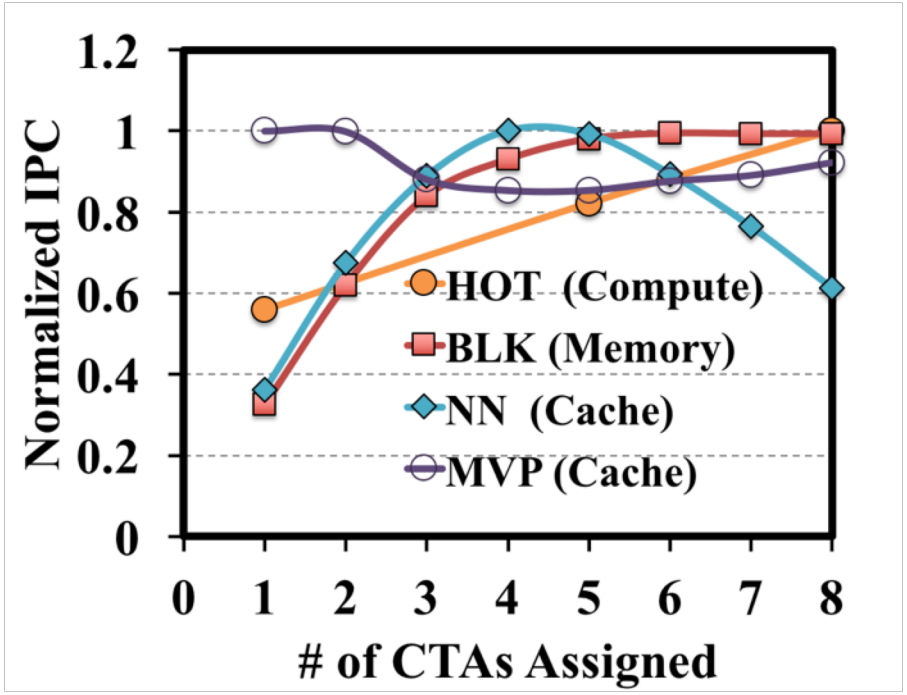
- Intelligently decide whether to choose between intra and inter SM slicing



Warped-slicer techniques

- Part 1 Kernel co-location optimization
- Part 2 Water-Filling algorithm
- Part 3 Dynamic profiling

Performance vs. CTA curve



- **Compute Intensive**:
  - More CTAs are better
- **Memory Intensive**:
  - Performance saturates quickly
- **L1 Cache Sensitive**:
  - Performance decreases when L1 cache is over-utilized.

Challenges: performance is NOT directly proportional to resources

Part 1: Kernel co-location optimization function

We propose to assign T_i thread blocks to kernel i by optimizing:

$$Max \ \ \min_{i} \ \ P(i, T_i)$$

So that: $$\sum_{i=1}^{K} R_{T_i} \leq R_{tot}$$

P: Performance (IPC)
R: Resources, including register files, shared memory, maximum threads and maximum # of thread blocks in each SM.

Naïve brute force algorithm

- Search all the combinations of allocations
- Assume maximum N CTAs from K kernels in an SM
- The brute force algorithm has the time complexity of

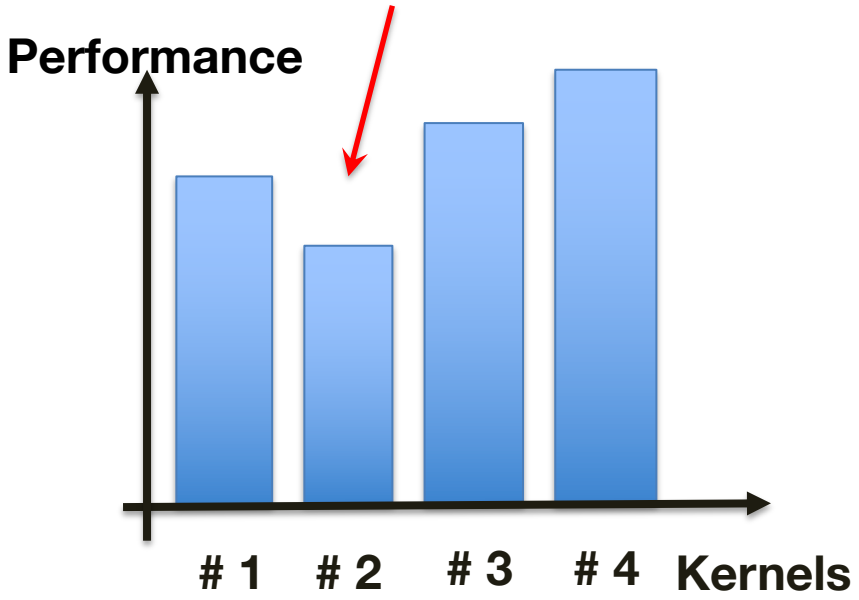
$$O(N^K)$$

Part 2: Water-filling algorithm

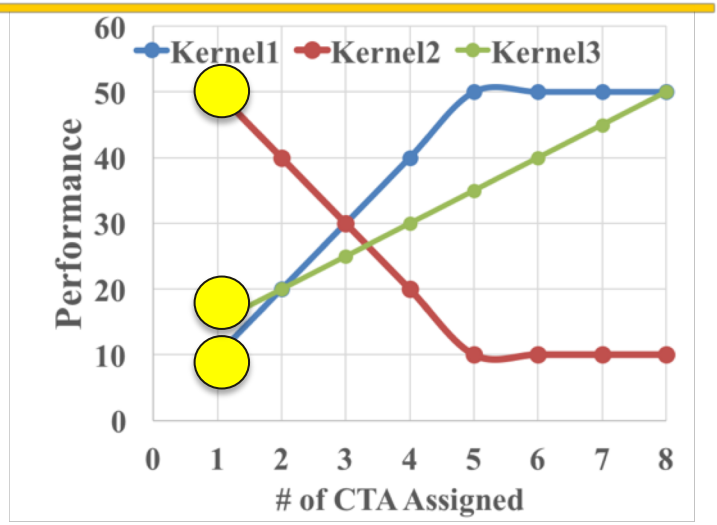
We propose an efficient algorithm to solve the MAX MIN problem

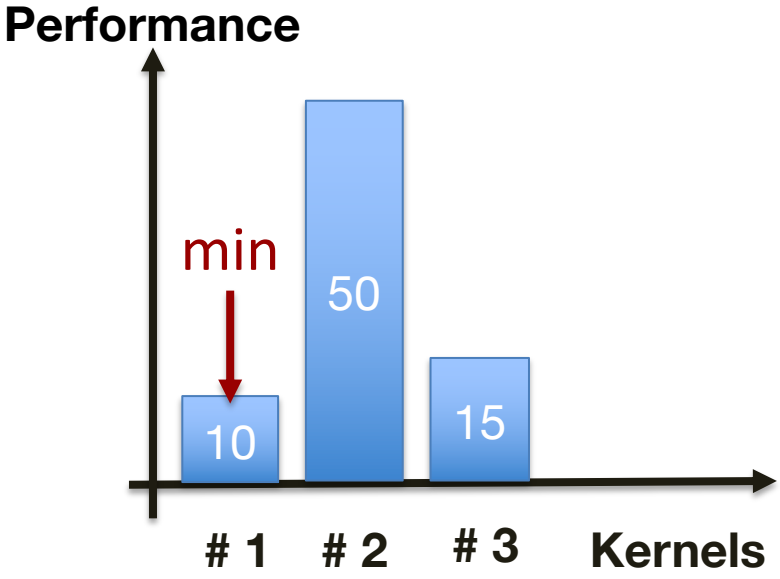
General idea:

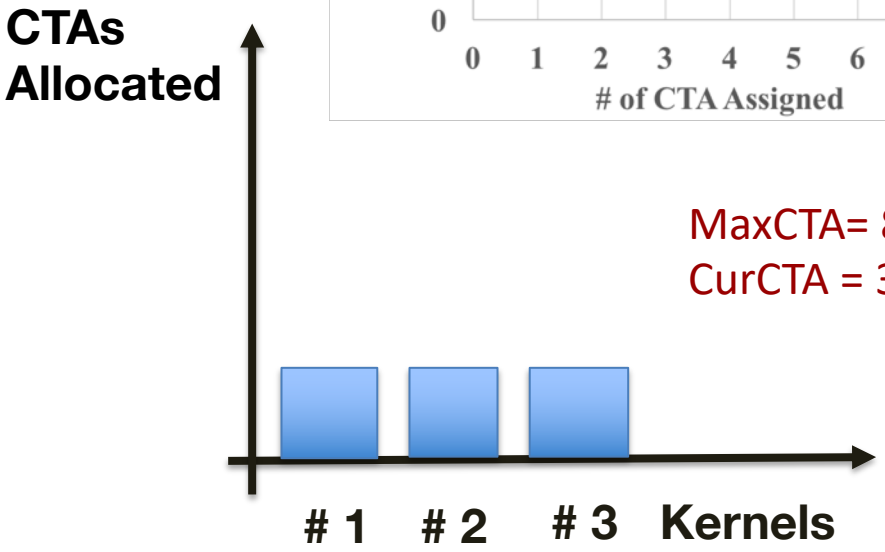
Greedily assign the minimum amount of the remaining resources to the kernel with the **MIN** performance so that the performance of that kernel increases



Part 2: Water-filling algorithm (iteration 1)

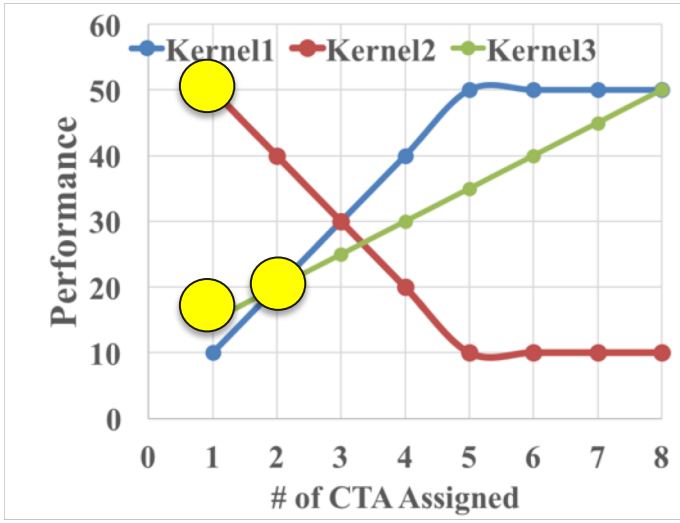


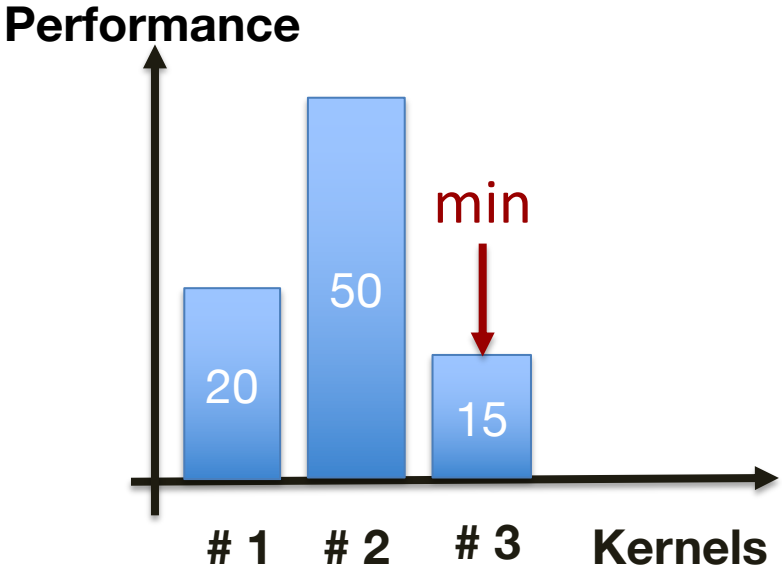


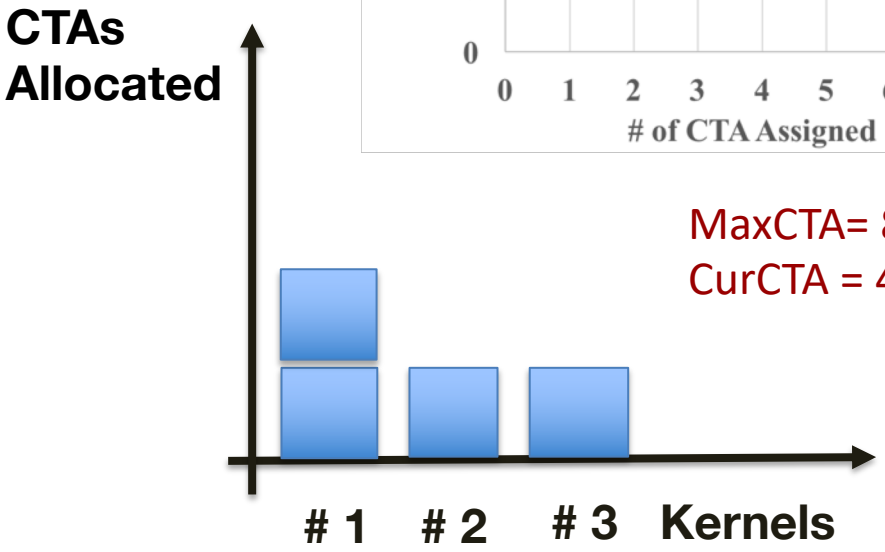


MaxCTA= 8
CurCTA = 3

Part 2: Water-filling algorithm (iteration 2)

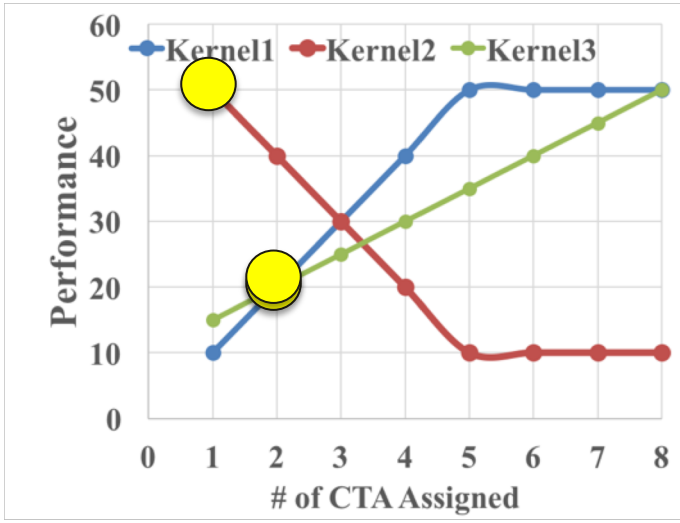


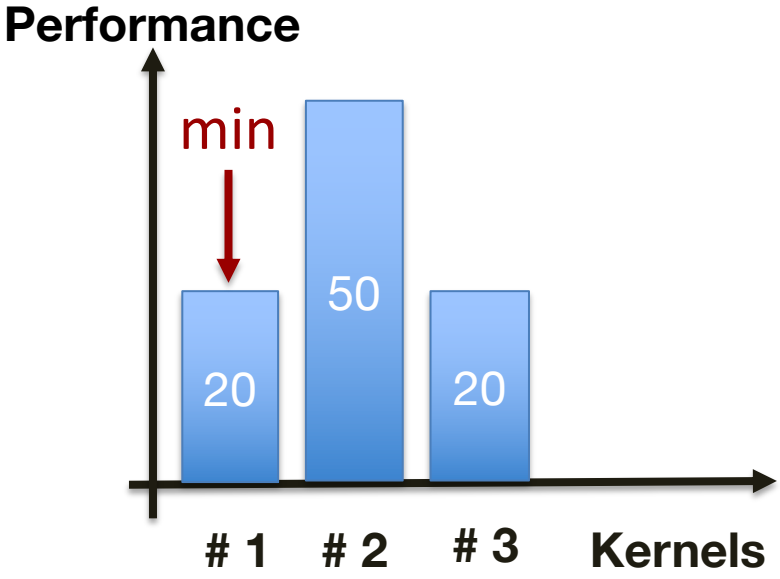


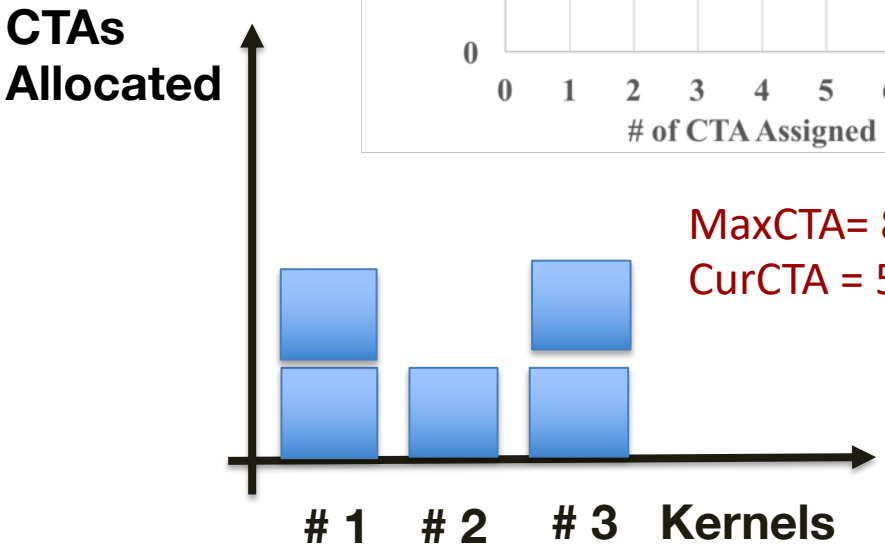


MaxCTA= 8
CurCTA = 4

Part 2: Water-filling algorithm (iteration 3)

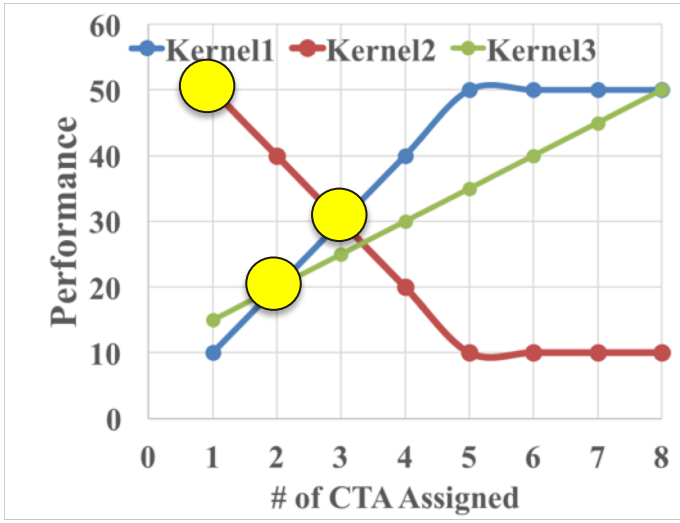


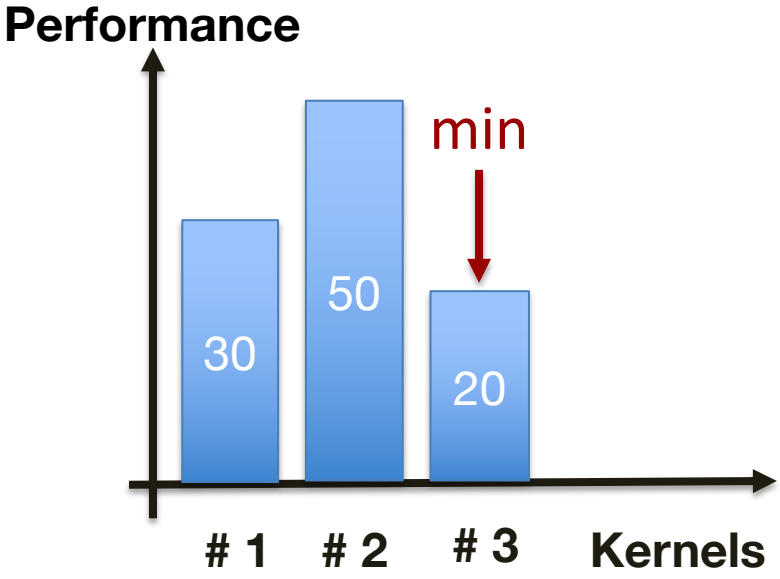


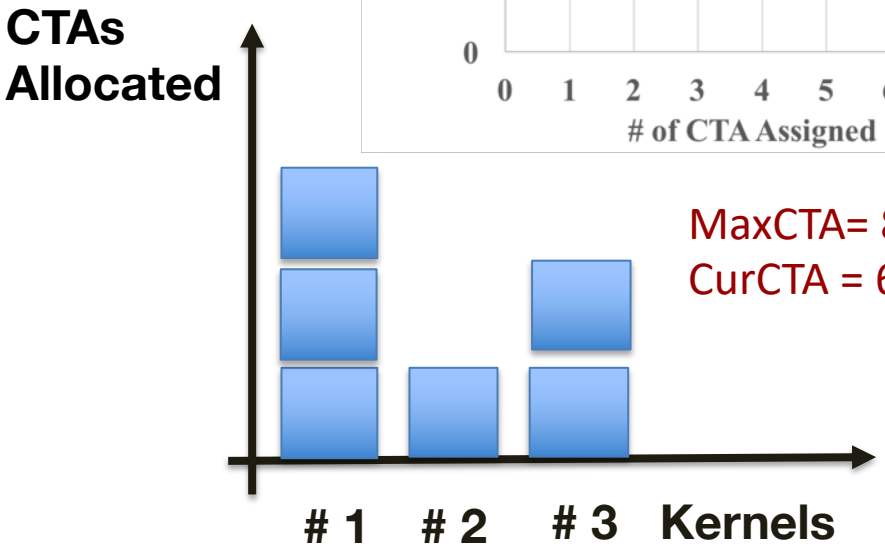


MaxCTA= 8
CurCTA = 5

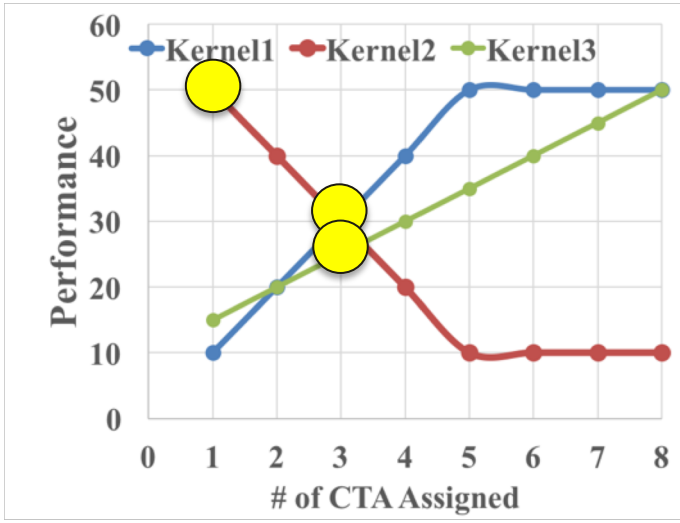
Part 2: Water-filling algorithm (iteration 4)

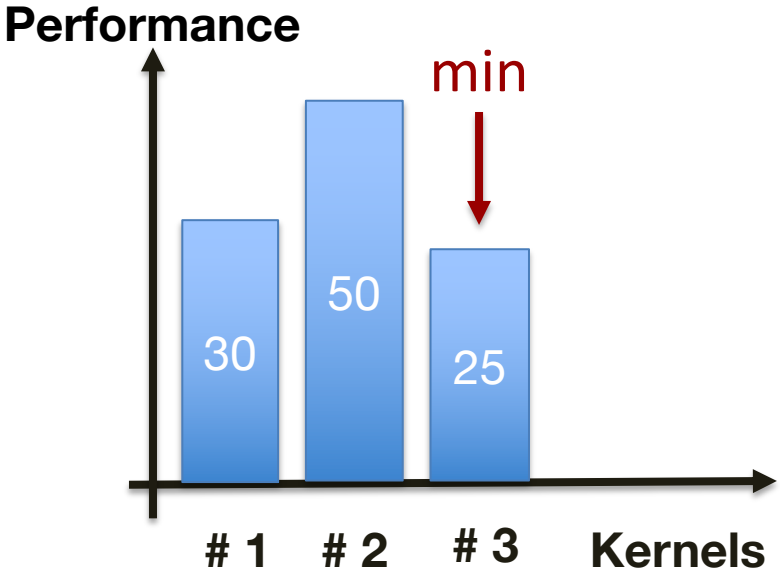


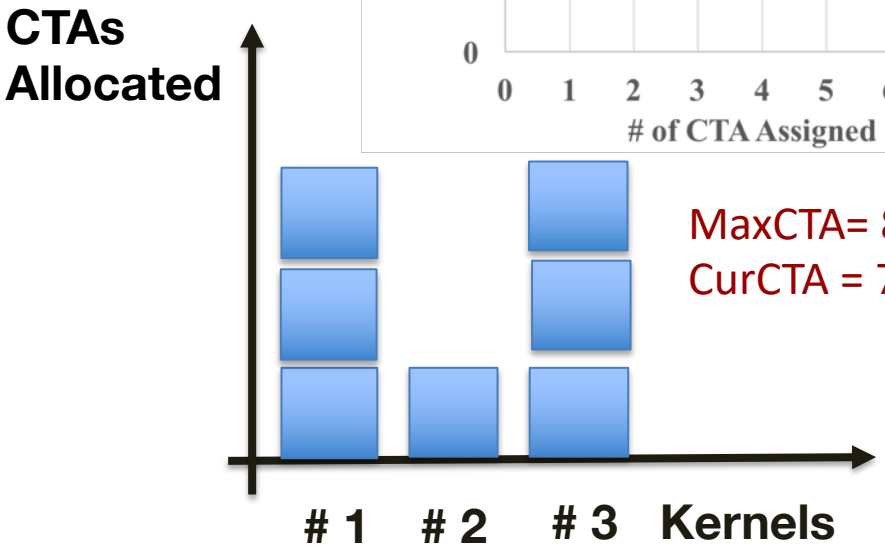




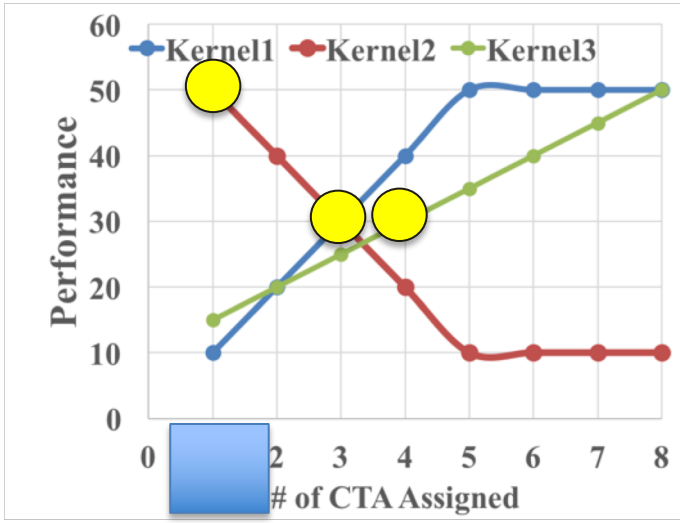
Part 2: Water-filling algorithm (iteration 5)

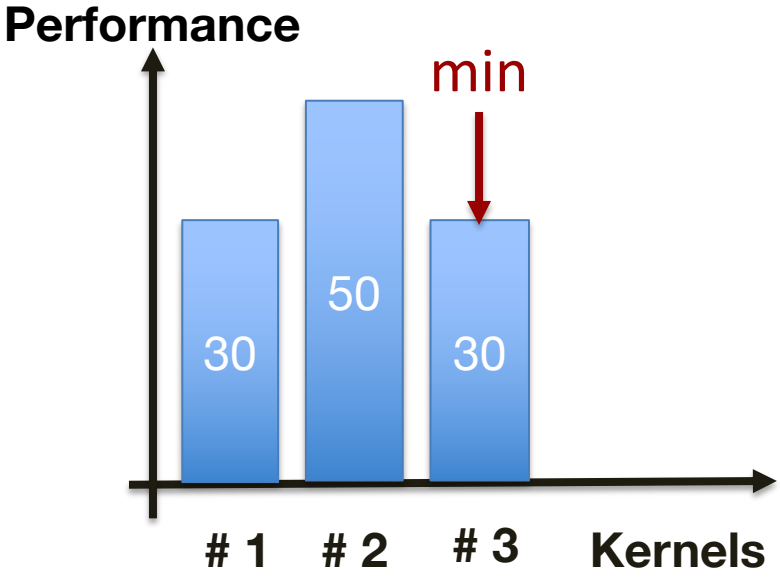


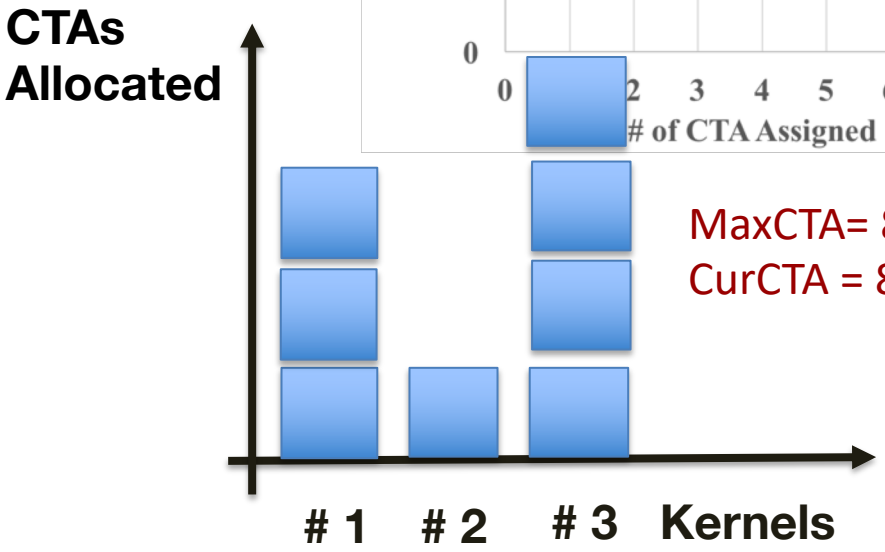




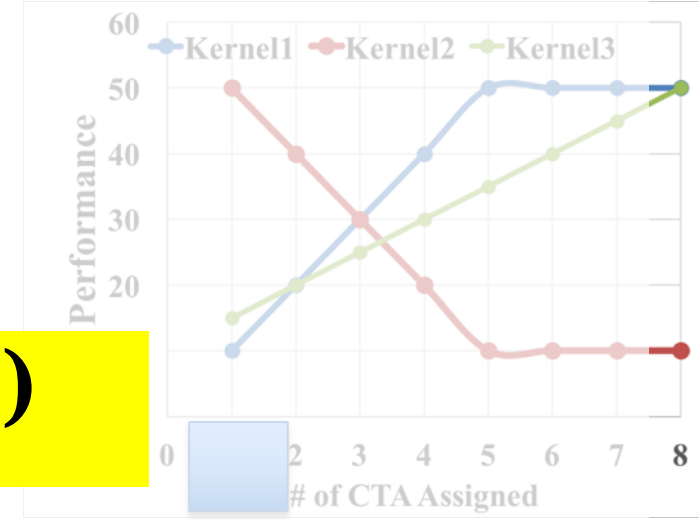
Part 2: Water-filling algorithm (iteration 6)



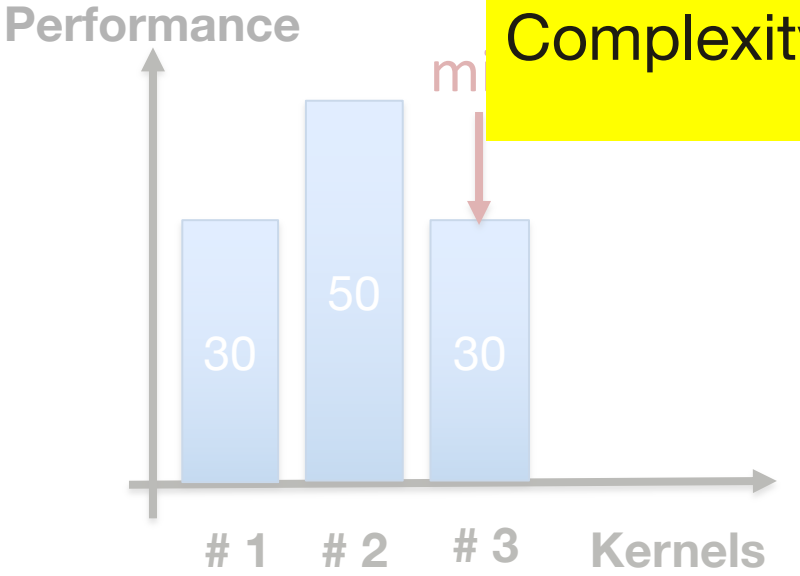


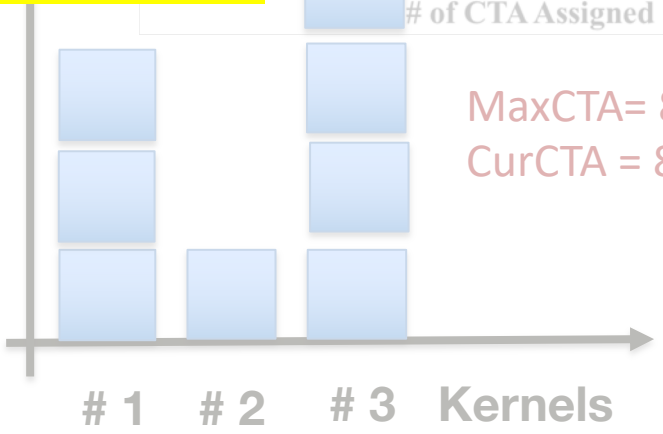


Part 2: Water-filling algorithm (iteration 6)



Complexity is $O(NK)$





Part 2: Water-filling algorithm

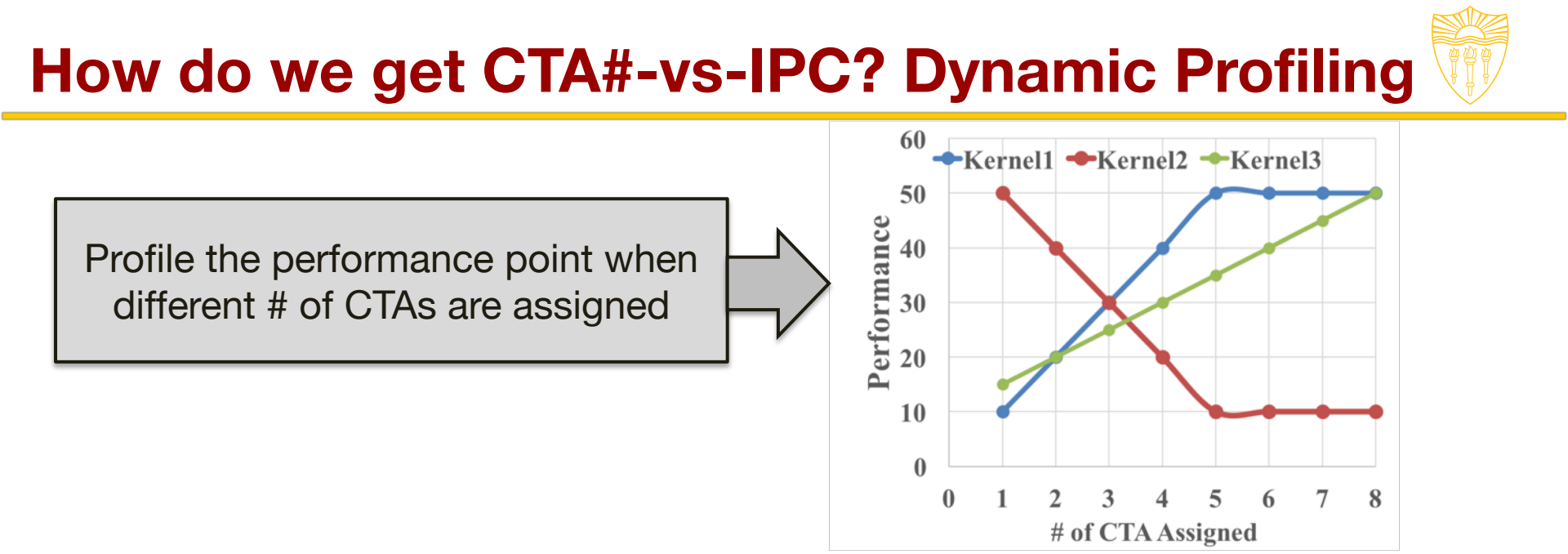
Limitation:

- Need to set performance loss upper bound

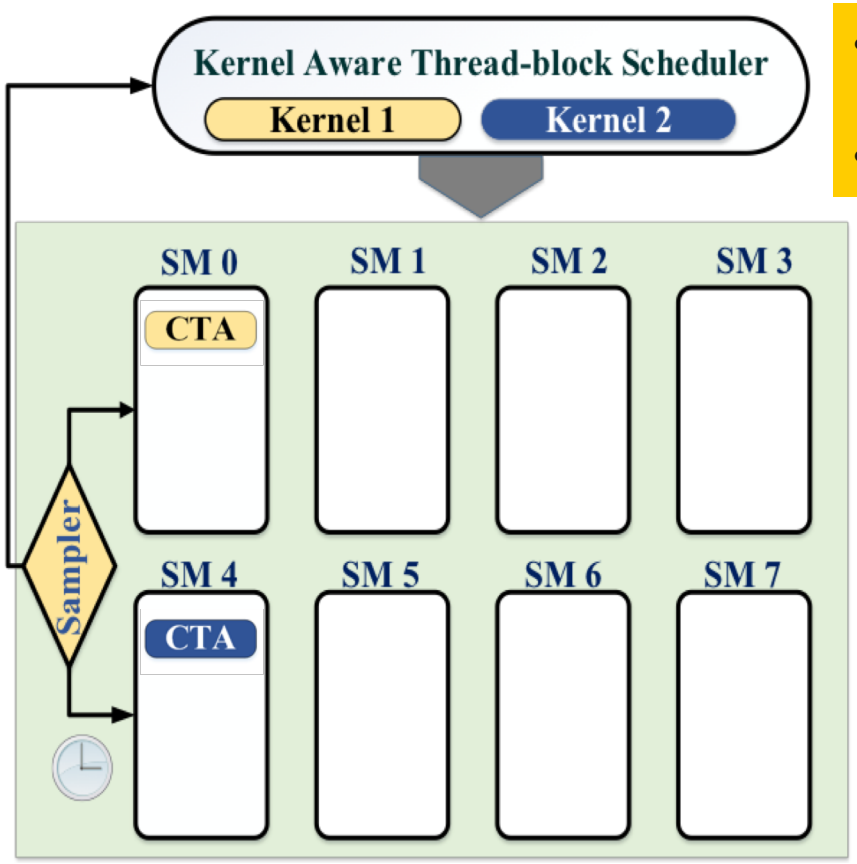
Solution:

- Set a threshold (60% for two kernels)
- ***Fall back on spatial multitasking*** when the performance loss exceeds the threshold.

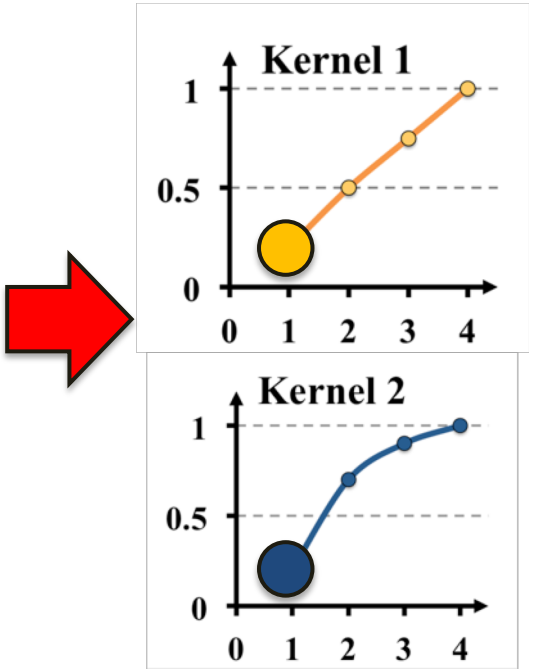
How do we get CTA#-vs-IPC? Dynamic Profiling



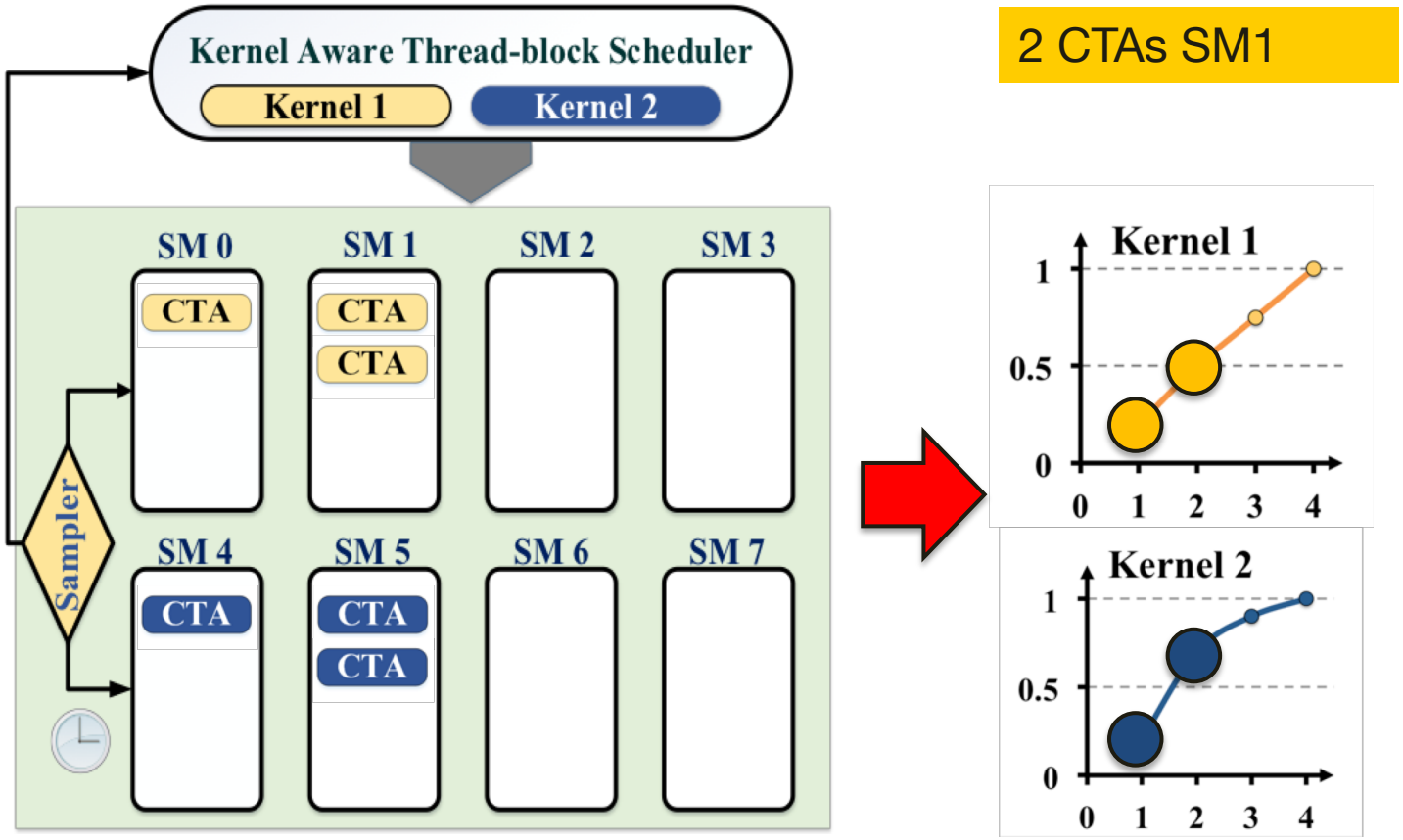
Part 3: Dynamic profiling



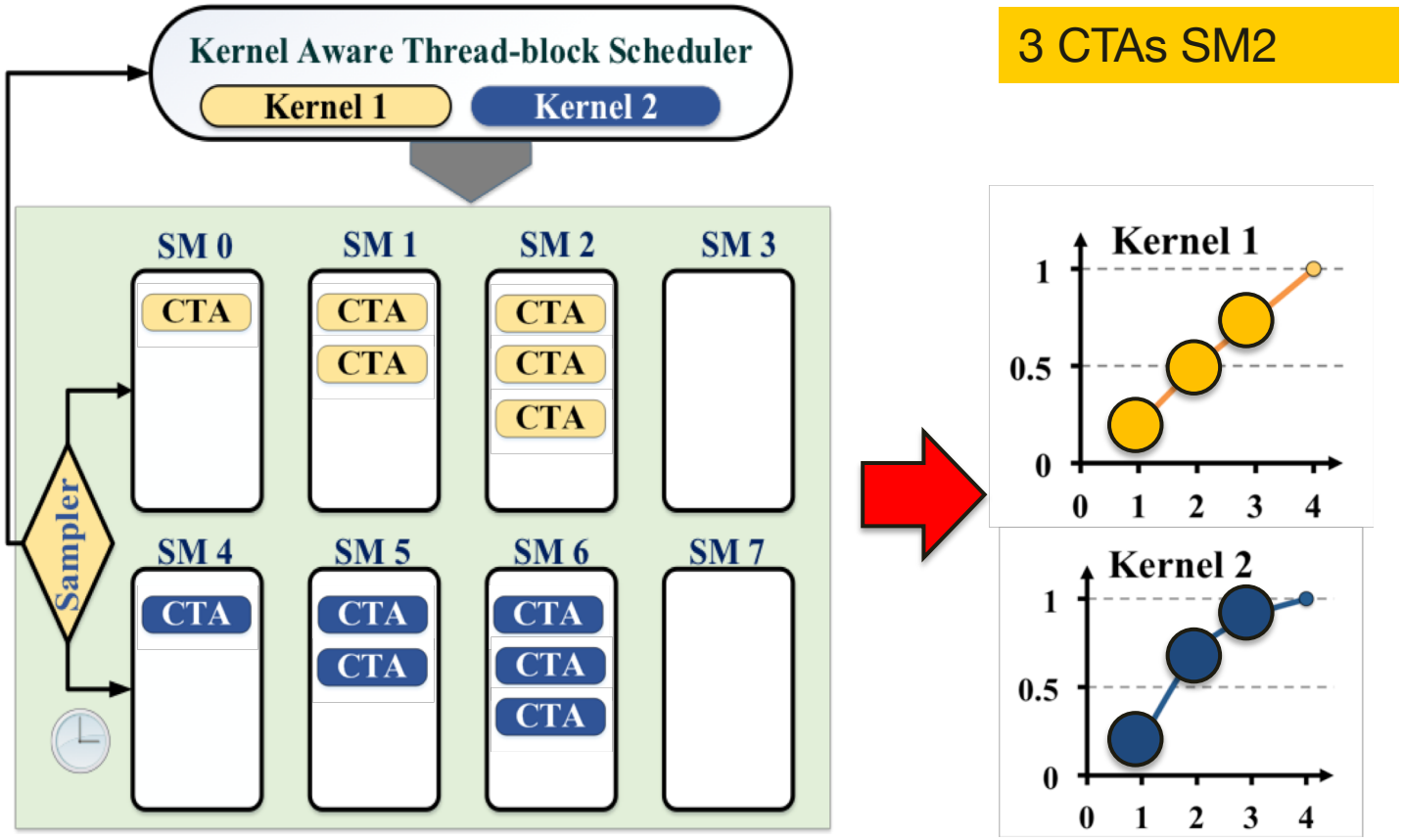
- We split SMs into two equal groups.
- We start with one CTA SM0.



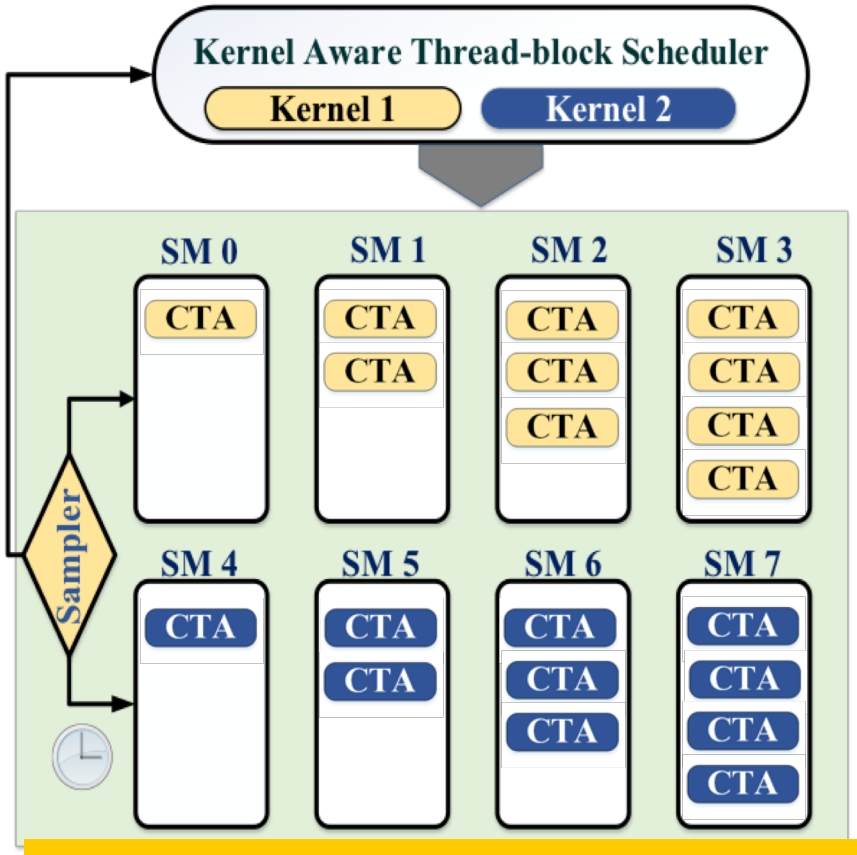
Part 3: Dynamic profiling



Part 3: Dynamic profiling

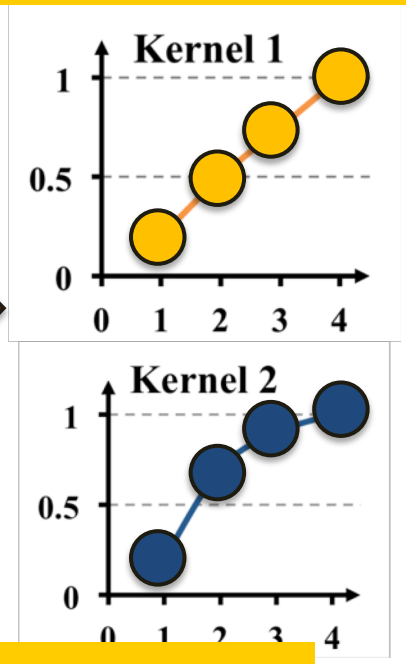


Part 3: Dynamic profiling



4 CTAs SM3

Now we can plot the two curves for the two kernels



For >2 kernels: time sharing one SM to run with a different number of CTAs sequentially

How to deal with interference?

Challenge:

The curve may be distorted:

- We assume independent accesses for each SM
- But in fact, the L2 and memory accesses are shared across all SM

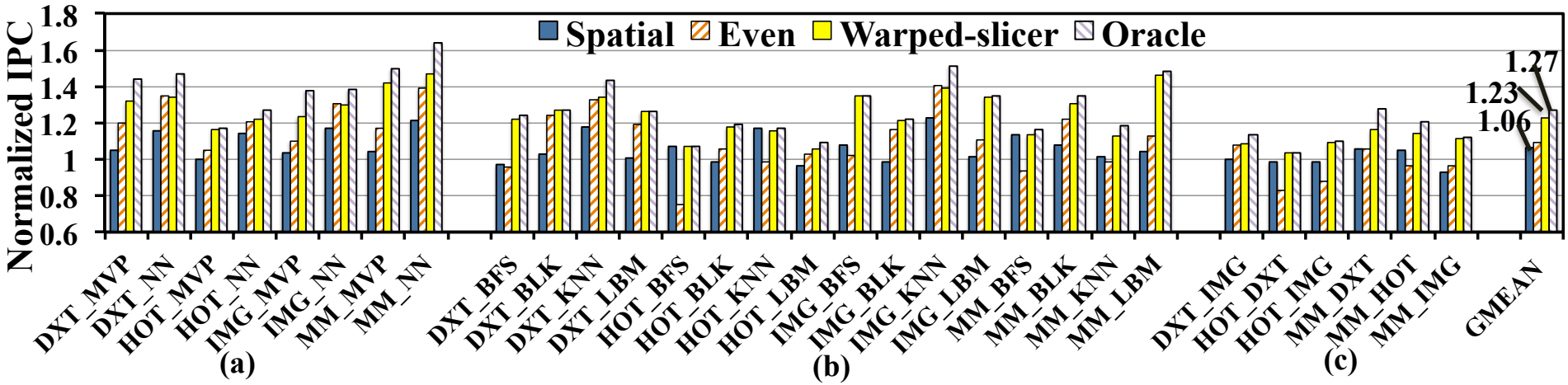
We design a scaling factor* to correct the curve.

*Please refer to detailed derivation in the paper.

Evaluation Methodology

- Simulating Environment:
  - GPGPU-Sim v3.2.2
  - Simulator front end is extensively modified to support multiprogramming
- Benchmarks:
  - 10 GPU applications from image processing, math, data mining, scientific computing and finance domains
- Experiments:
  - We kept the total instructions executed for each workload the same across different strategies
  - *Oracle partitioning*: acquired by **thousands** of experiments of all possible allocations

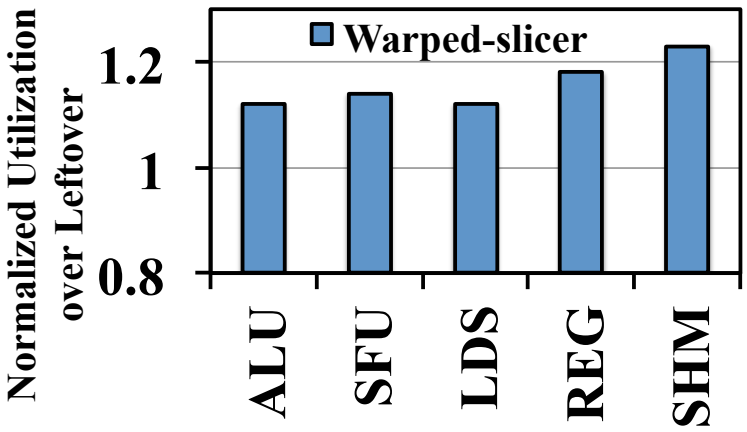
Performance result – 2 kernels



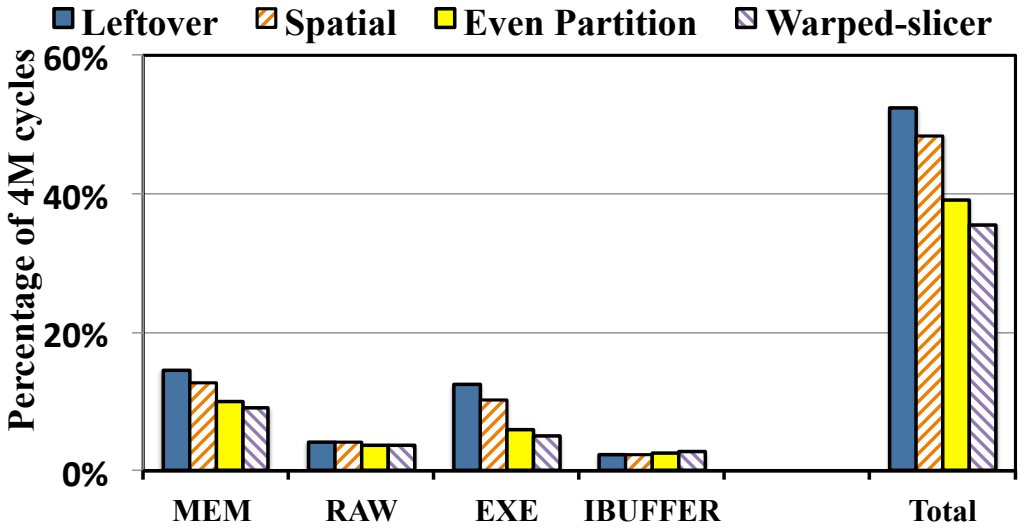
- Inter-SM (**Spatial**) 6% over **Leftover**
- **Warped-slicer** 23% better.
- **Warped-slicer** within 4% of Oracle

Resource utilization results – 2 kernels

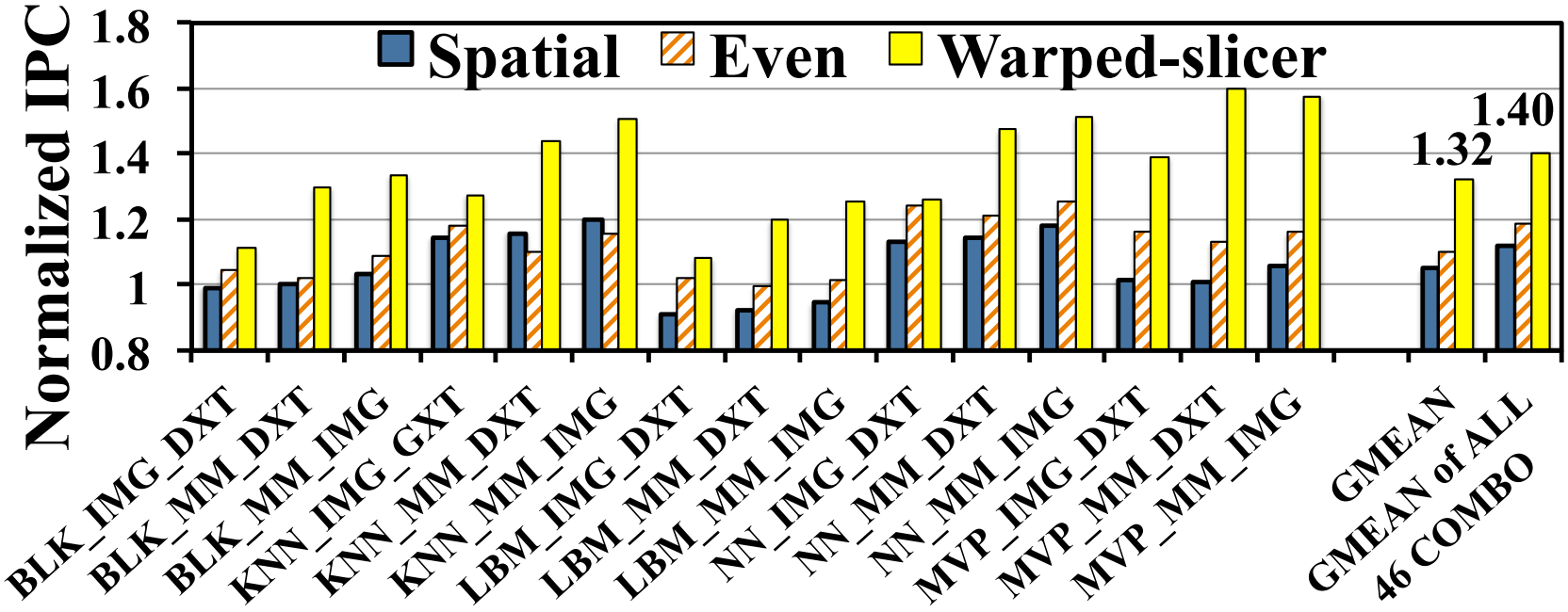
- 10-20% better resource utilization



- 18% reduction in stall cycles



Performance result – 3 kernels



Warped-slicer out-performs even partitioning by 21%

Summary

- We studied various intra-SM slicing techniques for GPU multiprogramming.
- We proposed **warped-slicer**, which achieves 23% performance improvement for 2 kernels and 40% performance improvement for 3 kernels over Leftover policy.
- We implemented the water-filling algorithm and various counters in Verilog. The hardware overhead is minimum.

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